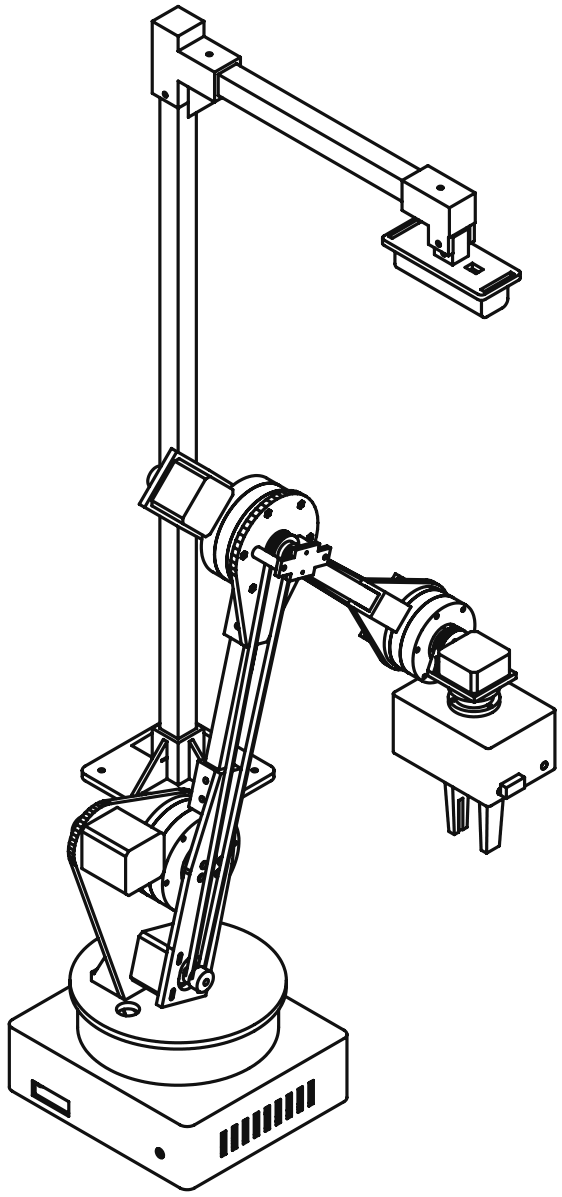
CHESS ROBOT ARM

Assembly instructions



36 steps. Read the whole step before you start it.

Drawings are generated from the CAD: what you see is what you print.

Parts in black are new in a step; grey parts are already in place.

Before you start

TOOLS

- [ ] Hex keys 1.5, 2, 2.5, 3 mm
- [ ] Small hammer or a vice (pins, bearings)
- [ ] Superglue (magnets)
- [ ] Bearing grease (ball race)
- [ ] Hacksaw + file (aluminium tube)
- [ ] Soldering iron, wire stripper
- [ ] Multimeter
- [ ] Pliers, side cutters

PRINT FIRST

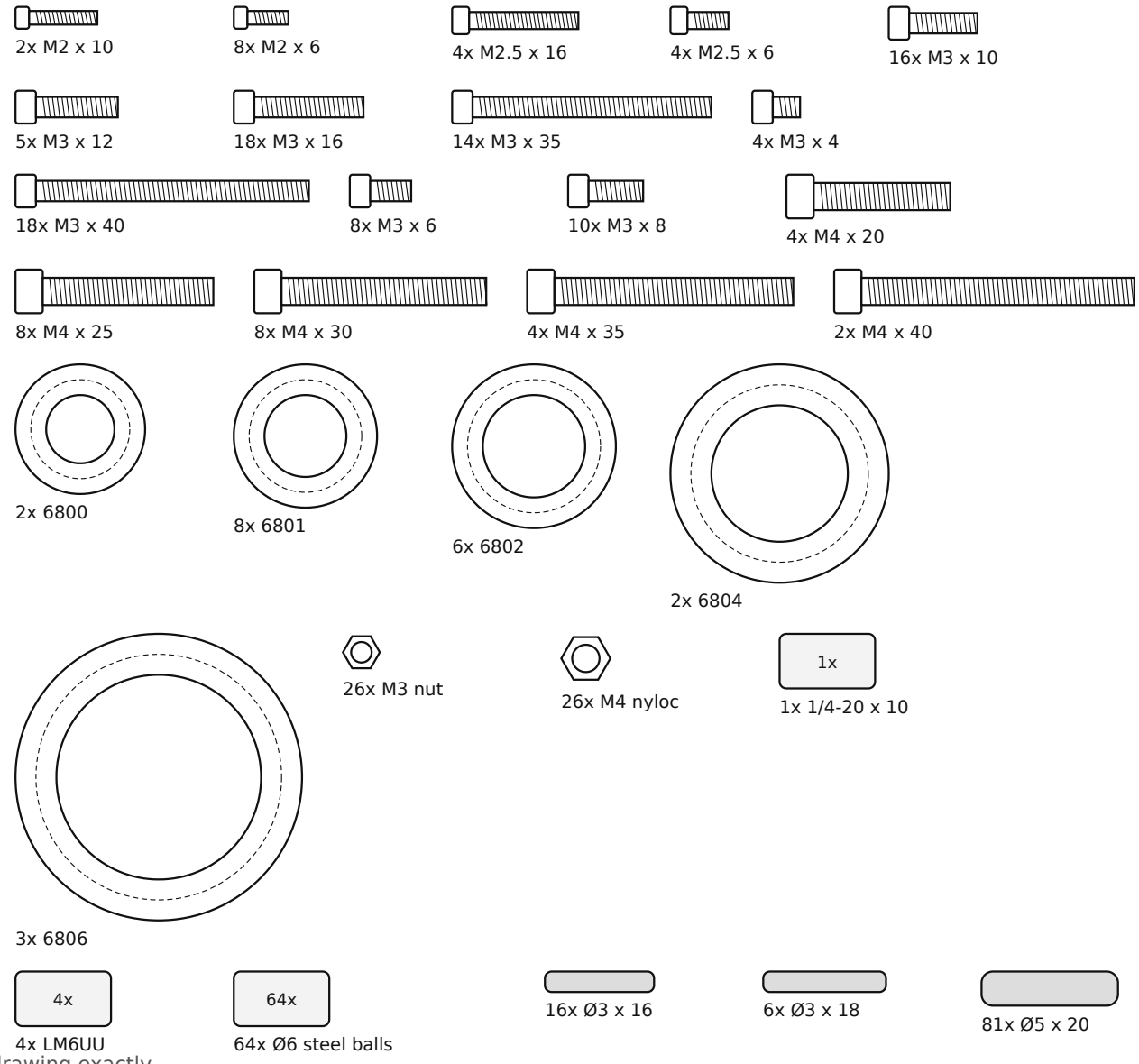
All parts: PETG, 0.2 mm, 4 walls.

Discs, pulleys, shafts, clamps, jaws, pinion: 100 % infill.

Everything else: 40 % infill.

See the drawing set for every part.

HARDWARE (all steps, actual size)

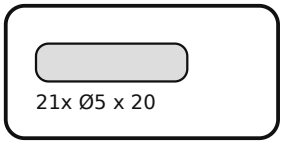


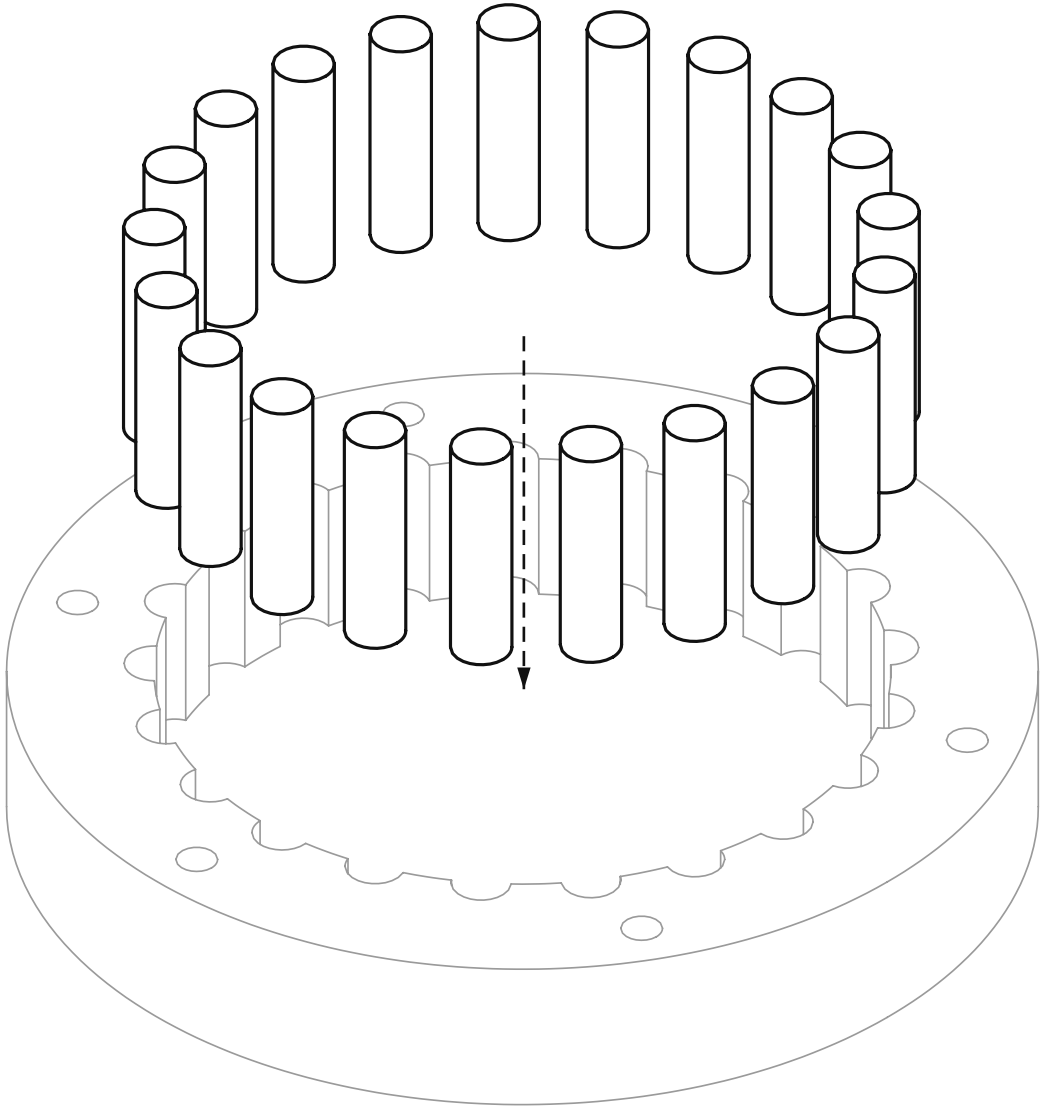
Check: print this page at 100 % and a screw should cover its drawing exactly.

1

Drive J1-J3: ring pins

Press the 21 dowels into the housing ring.
Tap them flush with the ring's top face.



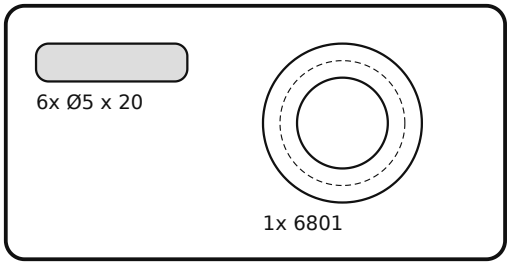


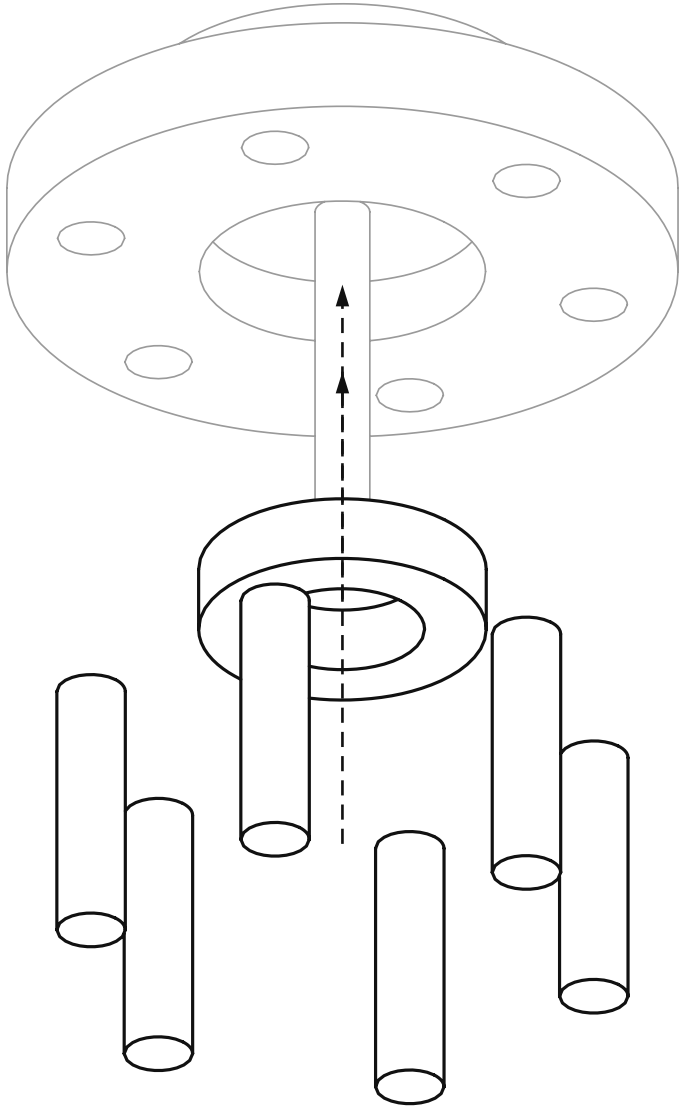
x3

2

Drive J1-J3: output flange

Press the output pins into the flange, then the small bearing into the flange's centre seat (from the pin side).



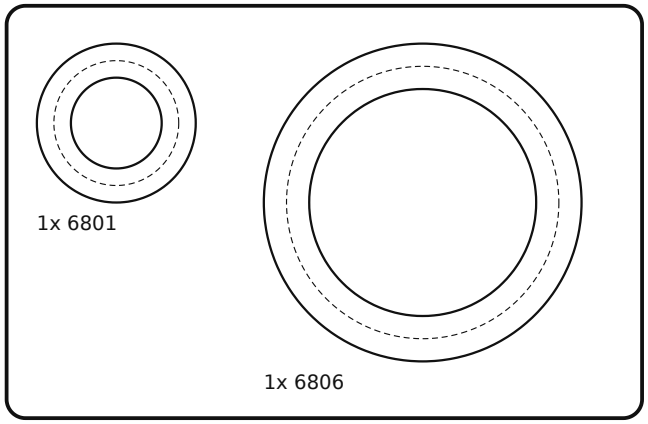


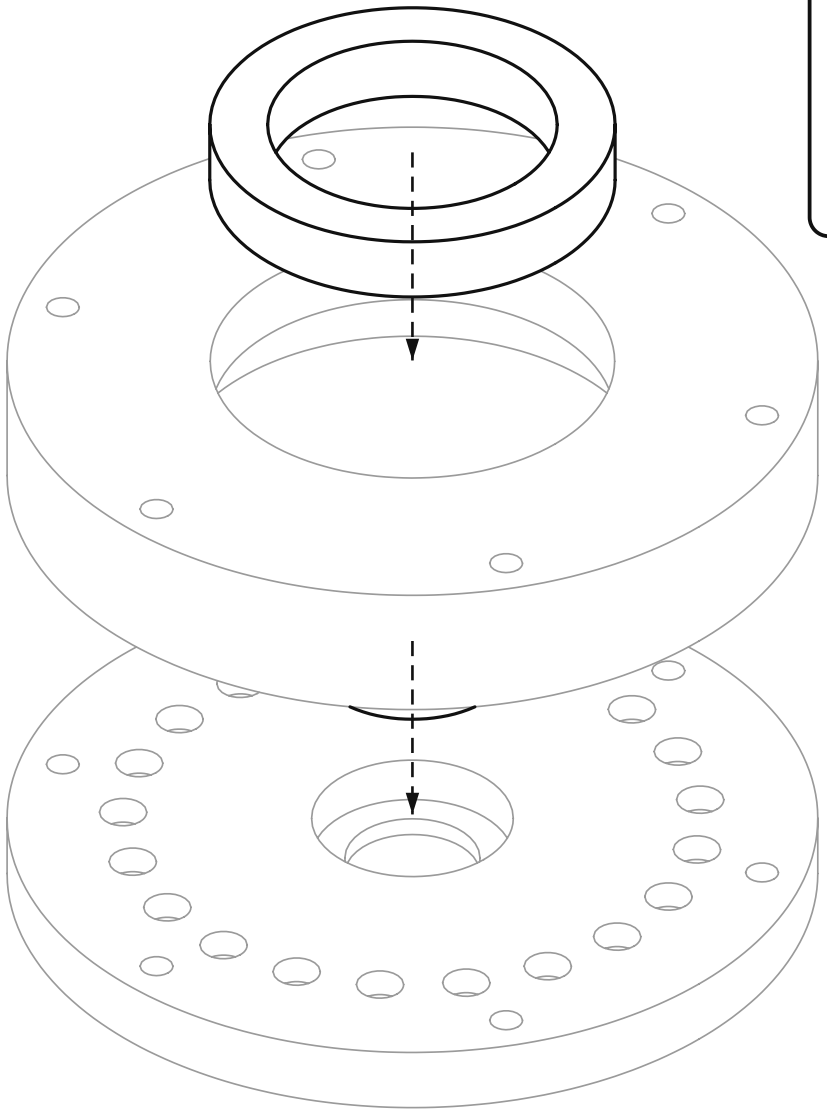
x3

3

Drive J1-J3: bearings in cover and cap

Press one small bearing into the back cover and the big bearing into the output cap. Push on the outer ring only.



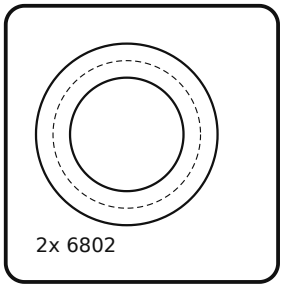


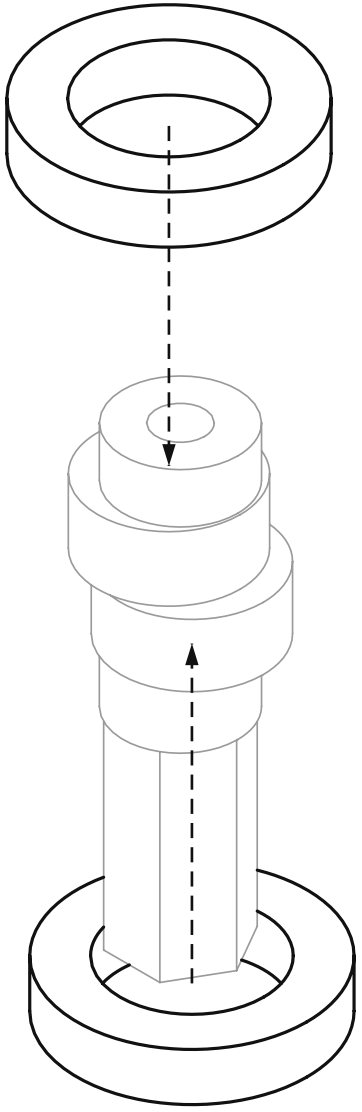
x3

4

Drive J1-J3: input shaft

Slide one bearing over each eccentric: the first from the pulley end, the second from the other end.



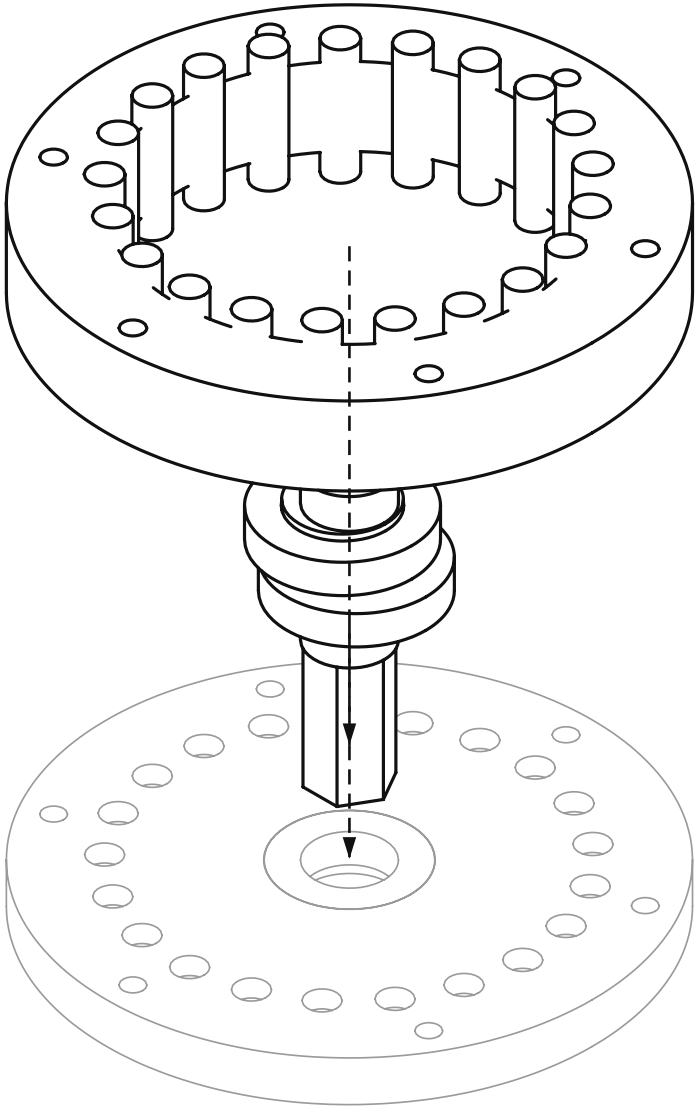


x3

5

Drive J1-J3: cover, shaft, ring

Shaft into the cover bearing, then lower the ring with its pins into the cover's pin holes.

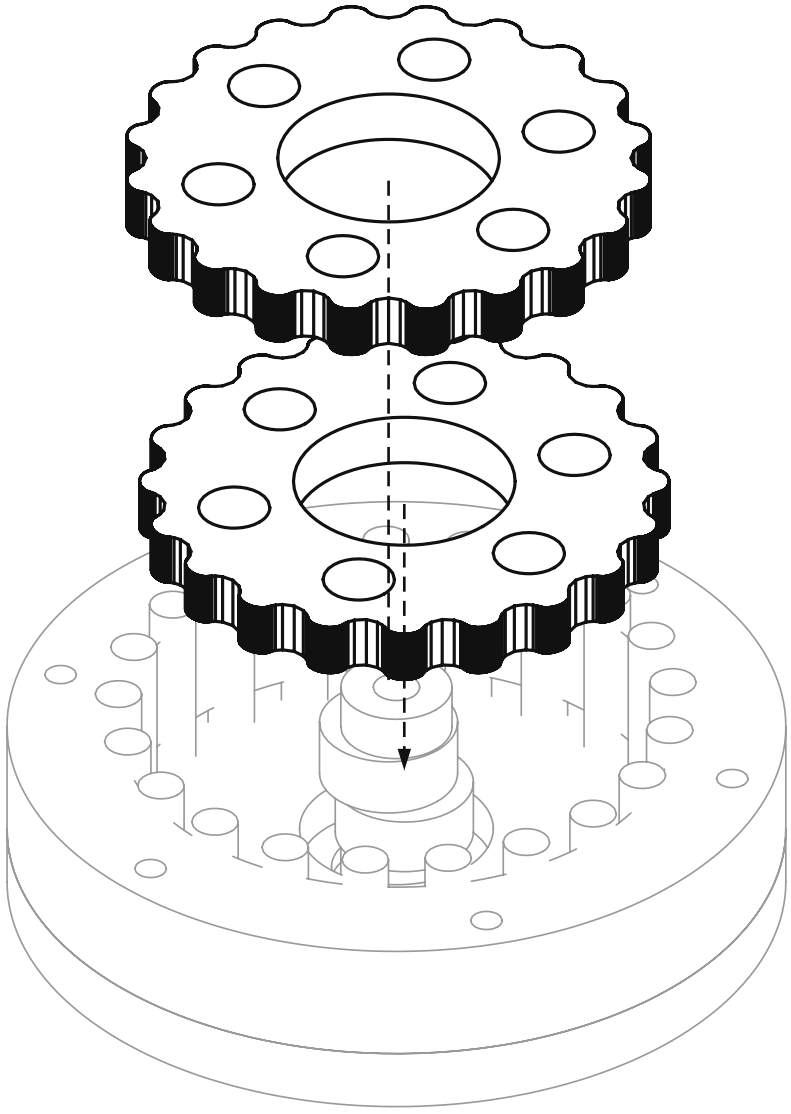


x3

6

Drive J1-J3: discs

Drop in disc A, then disc B. Turn the shaft a little until each disc falls flat between the pins.

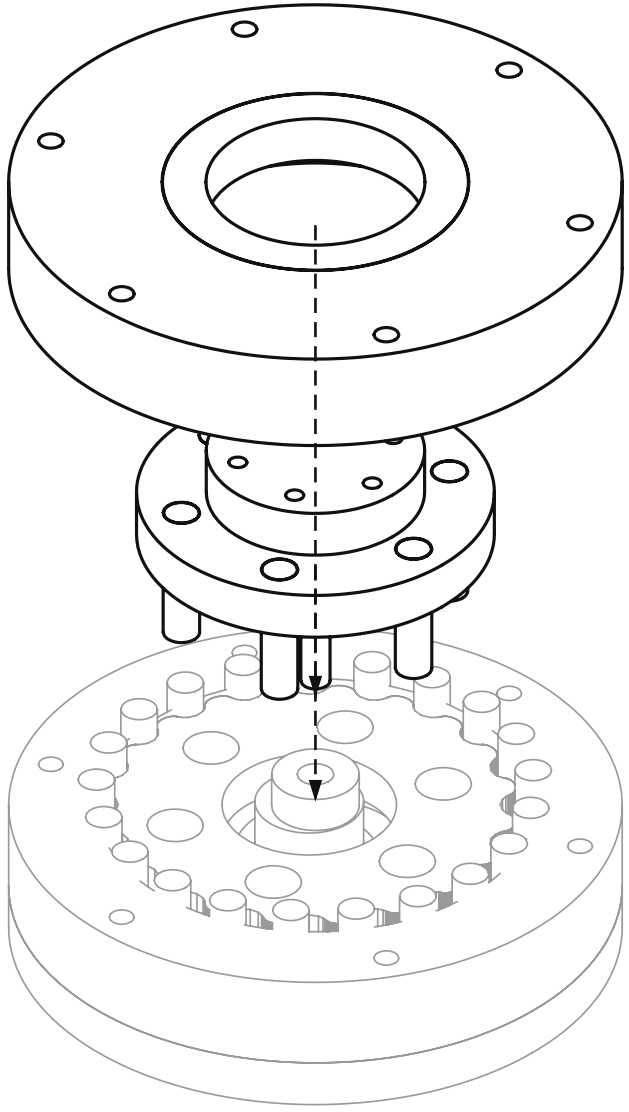


x3

7

Drive J1-J3: output flange and cap

Feed the flange pins through both discs, onto the shaft's front journal.
Cap on top. Keep the stack together until it is bolted down.

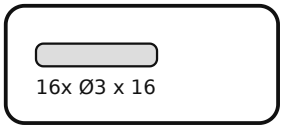


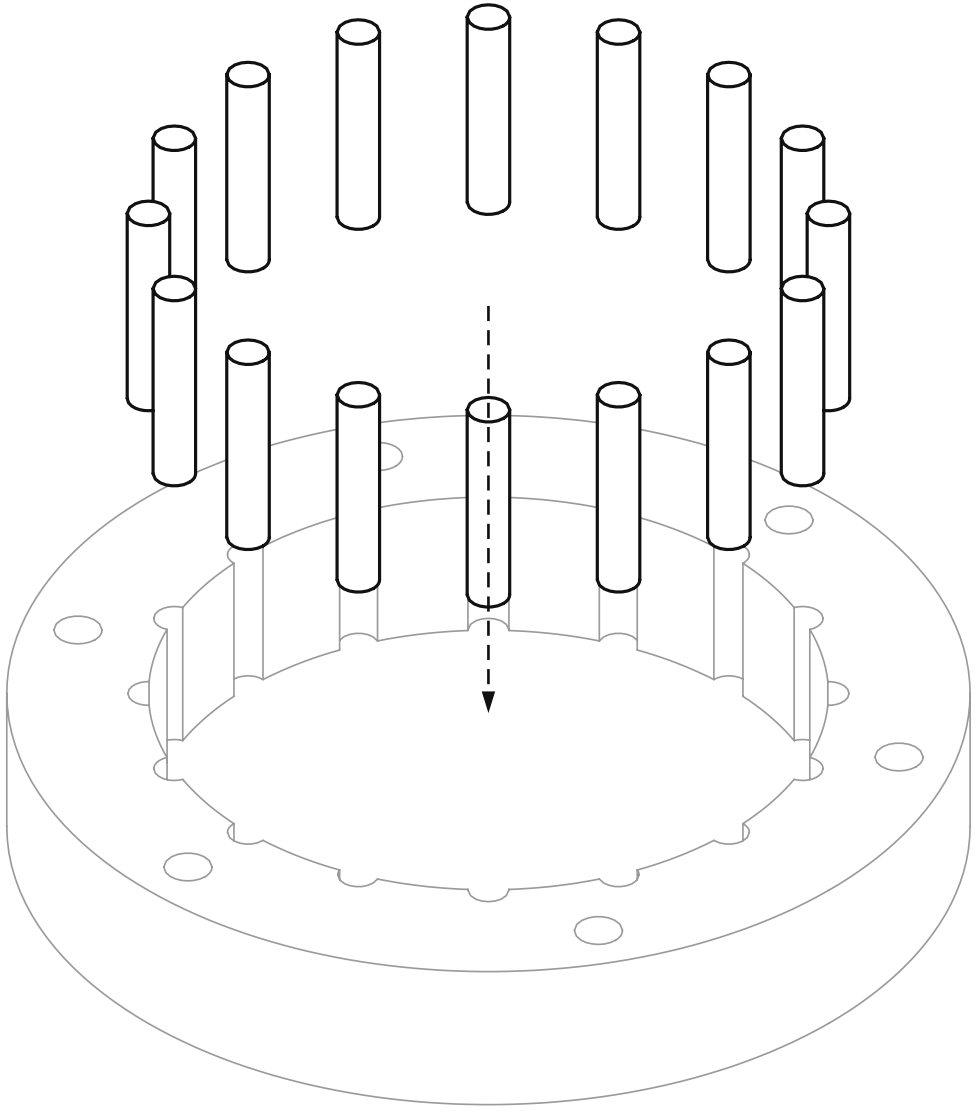
x3

8

Drive J4: ring pins

Press the 16 dowels into the housing ring.
Tap them flush with the ring's top face.



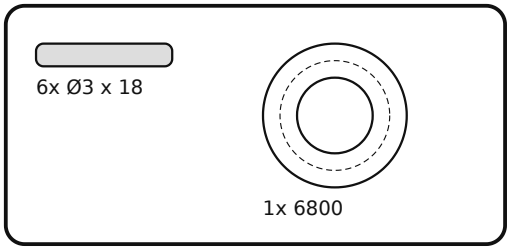


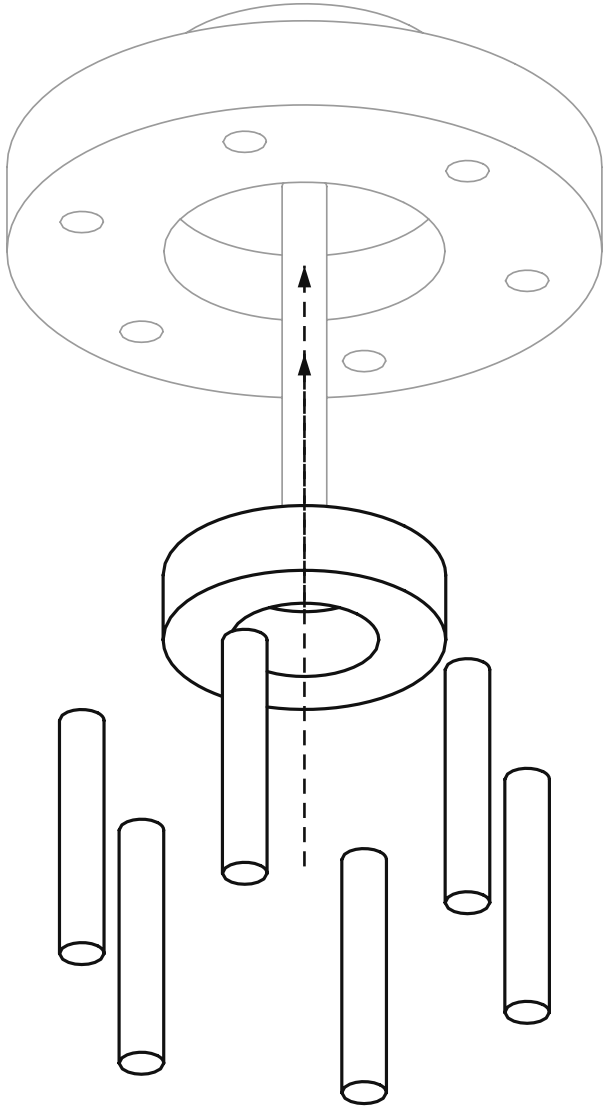
x1

9

Drive J4: output flange

Press the output pins into the flange, then the small bearing into the flange's centre seat (from the pin side).

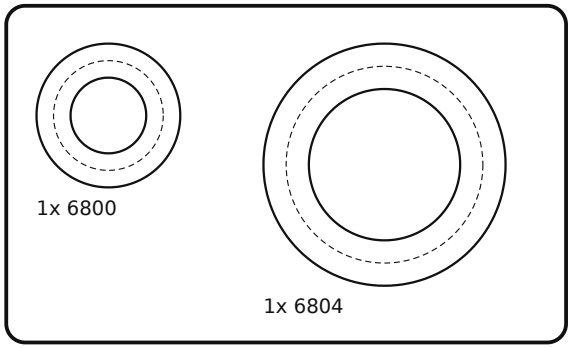


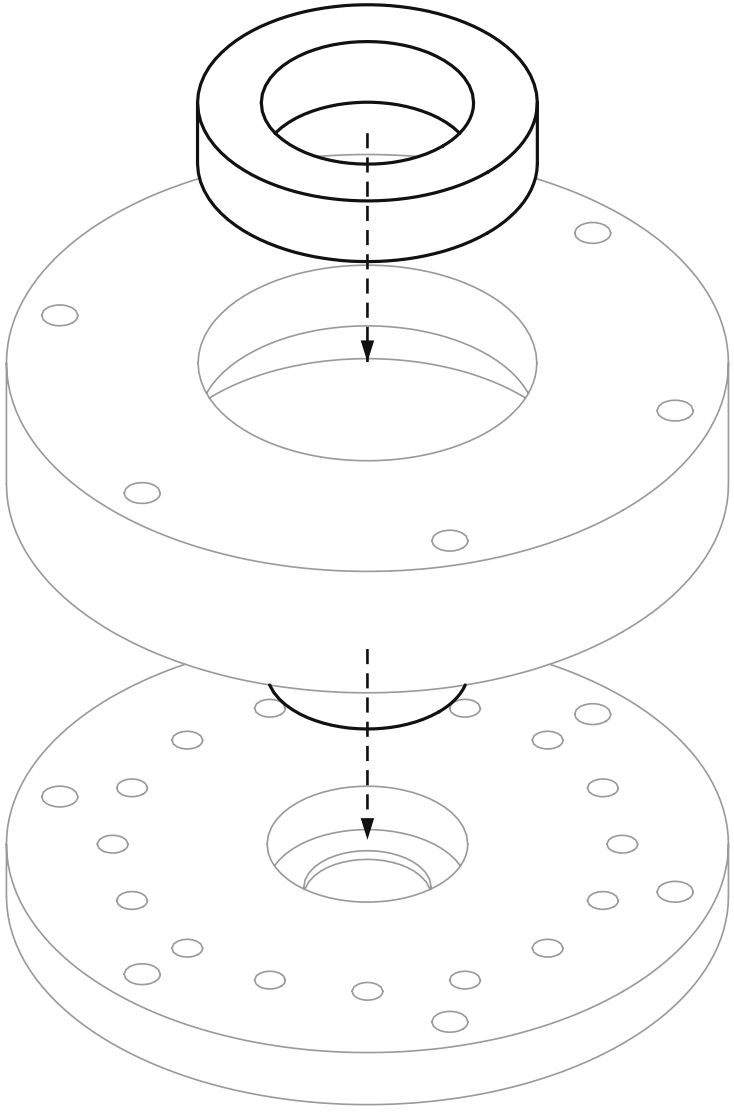


x1

10 Drive J4: bearings in cover and cap

Press one small bearing into the back cover and the big bearing into the output cap. Push on the outer ring only.

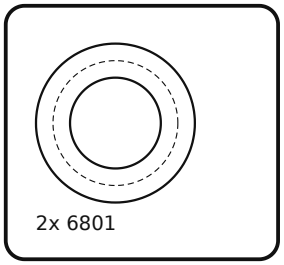


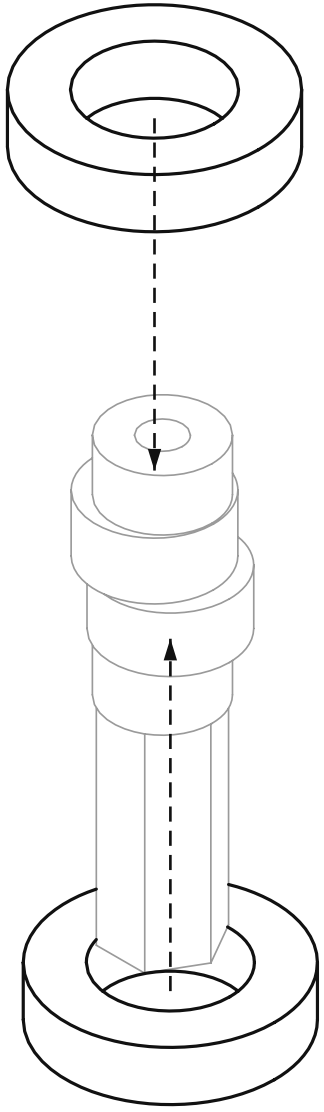


x1

11 Drive J4: input shaft

Slide one bearing over each eccentric: the first from the pulley end, the second from the other end.

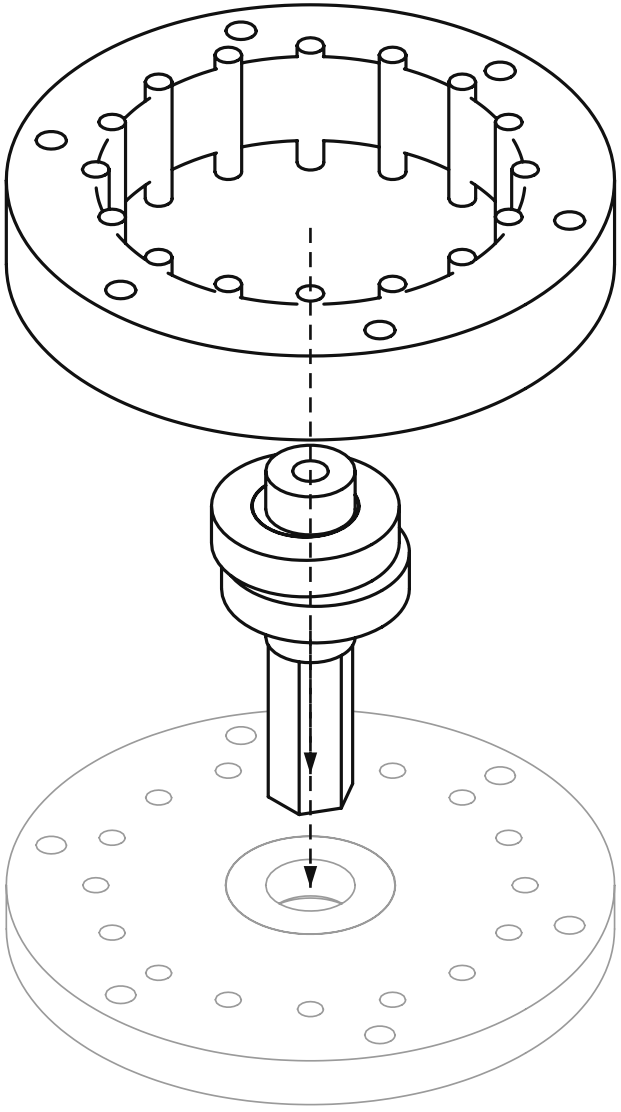




x1

12 Drive J4: cover, shaft, ring

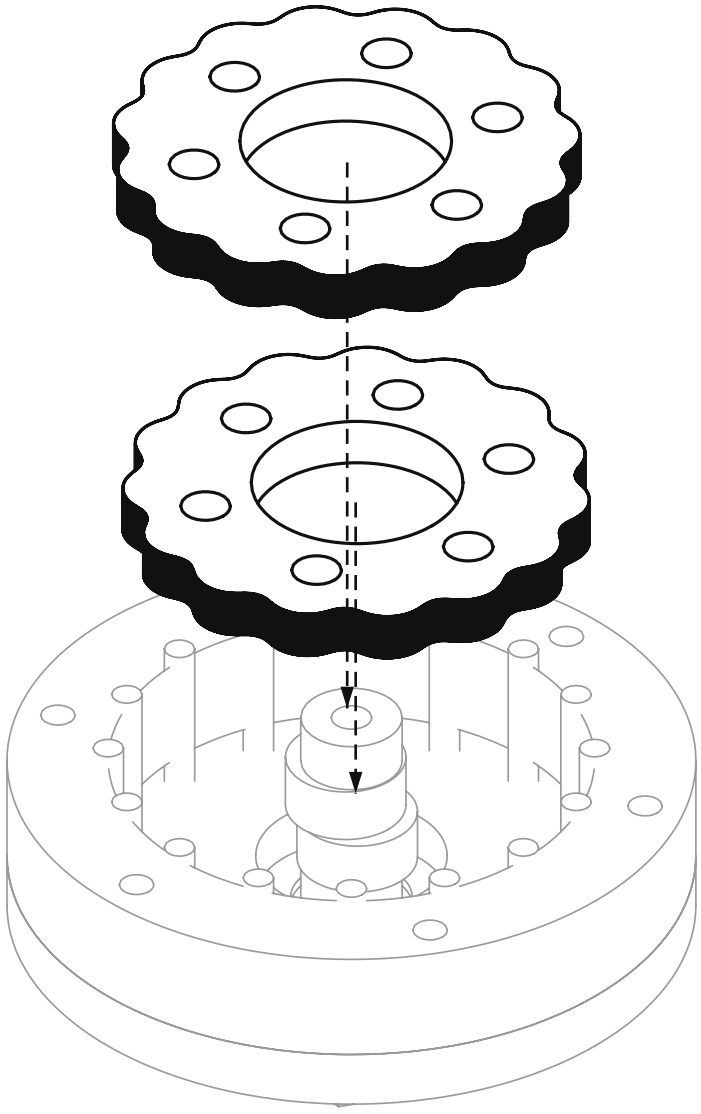
Shaft into the cover bearing, then lower the ring with its pins into the cover's pin holes.



x1

13 Drive J4: discs

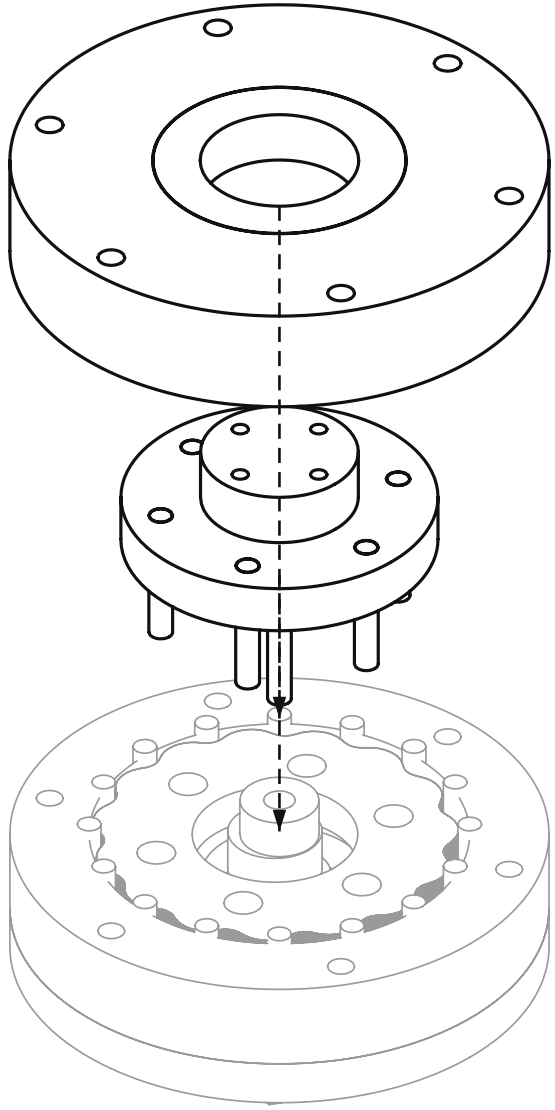
Drop in disc A, then disc B. Turn the shaft a little until each disc falls flat between the pins.



x1

14 Drive J4: output flange and cap

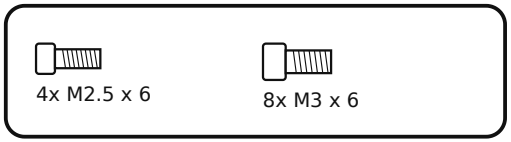
Feed the flange pins through both discs, onto the shaft's front journal.
Cap on top. Keep the stack together until it is bolted down.

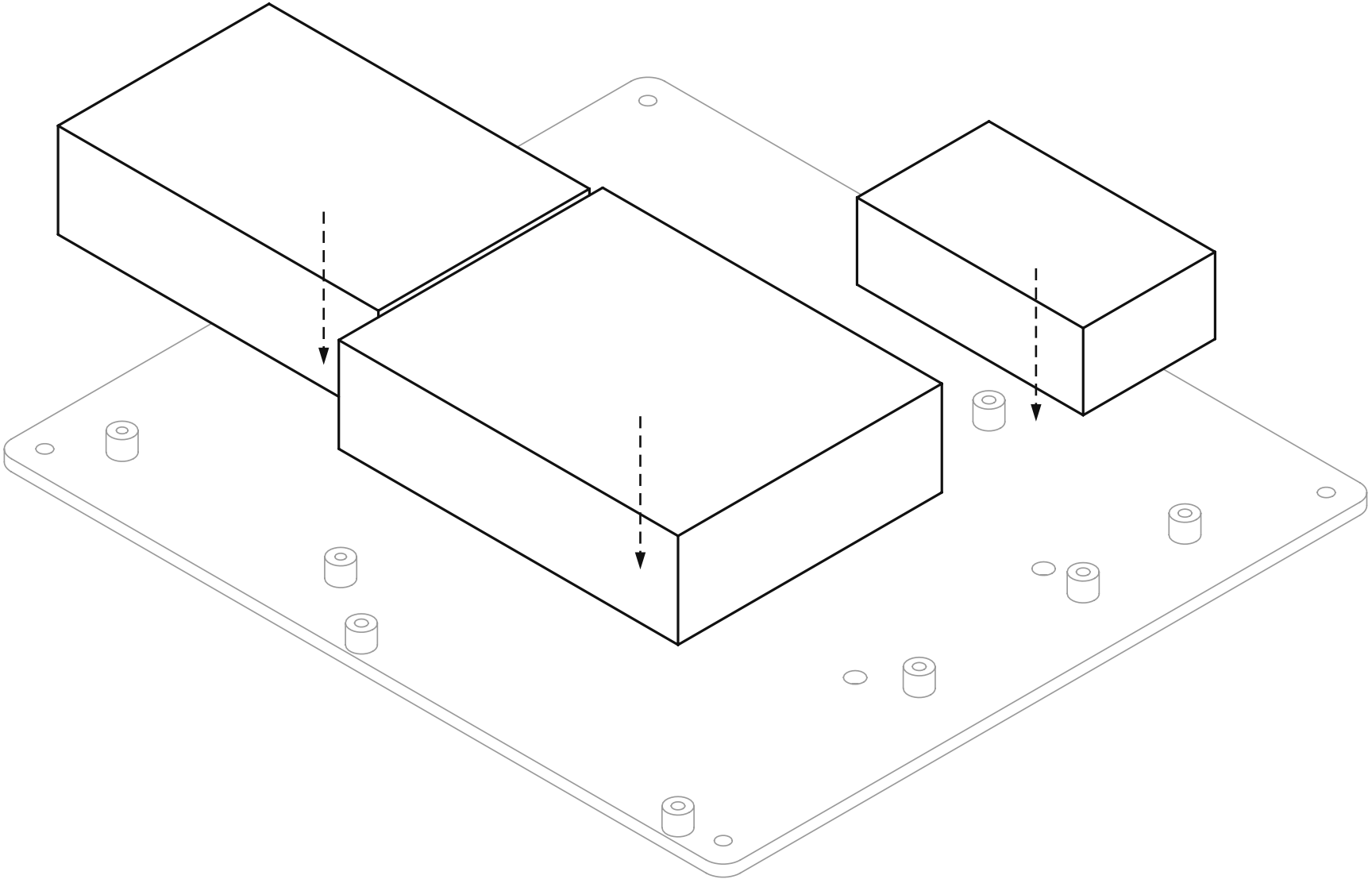


x1

15 Base: electronics on the floor

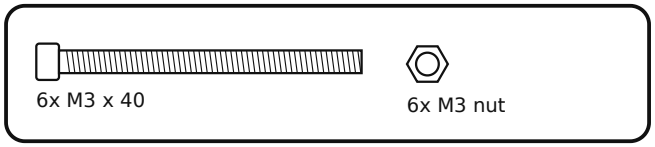
Pi 4, the controller perfboard and the buck converter go on their standoffs.

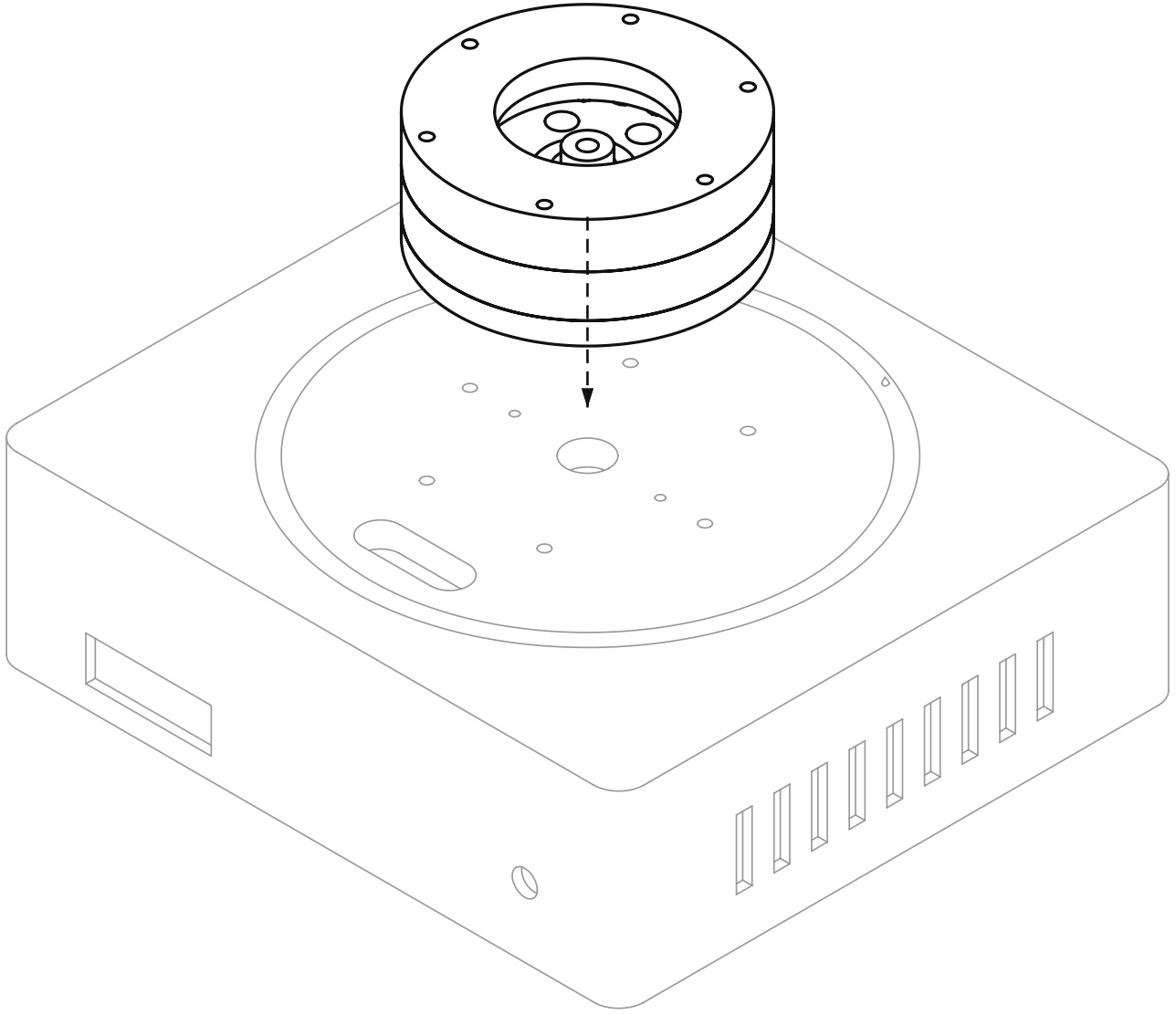




16 Base: J1 drive

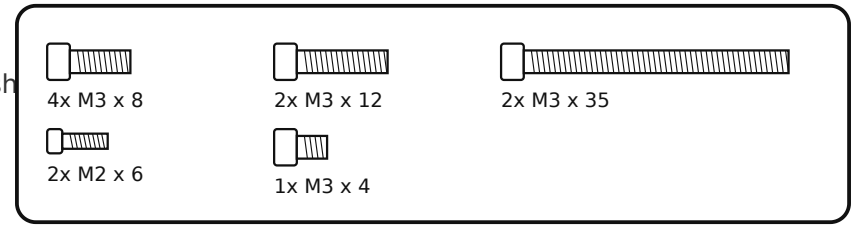
Drive on top of the base, bolts from the cap side, nuts in the pockets under the top plate.

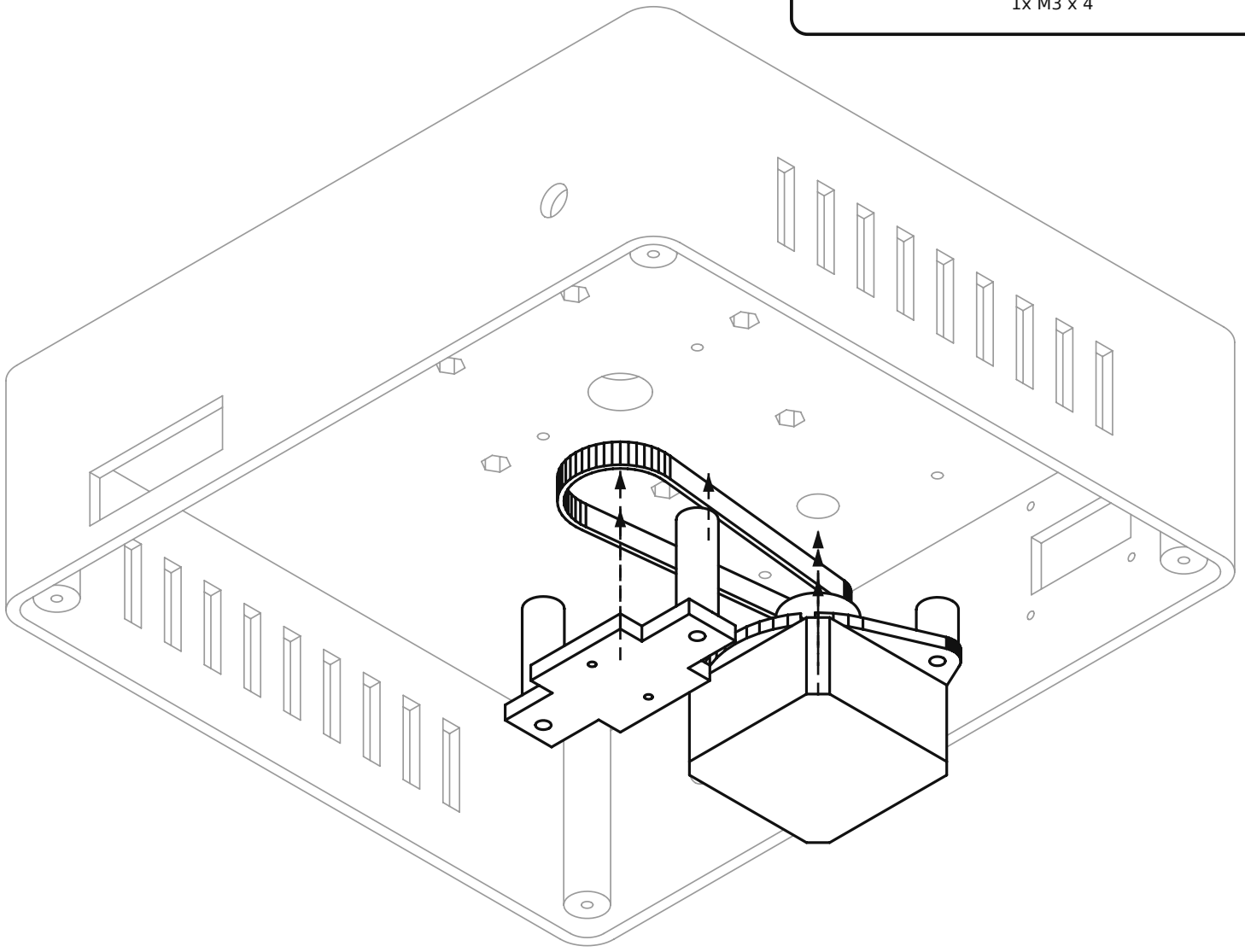




17 Base: J1 belt and sensor (from below)

Pulley onto the shaft hex (grub screw), magnet cap onto the rod, motor on its sh
belt on, shelf up onto its posts, sensor bridge last.

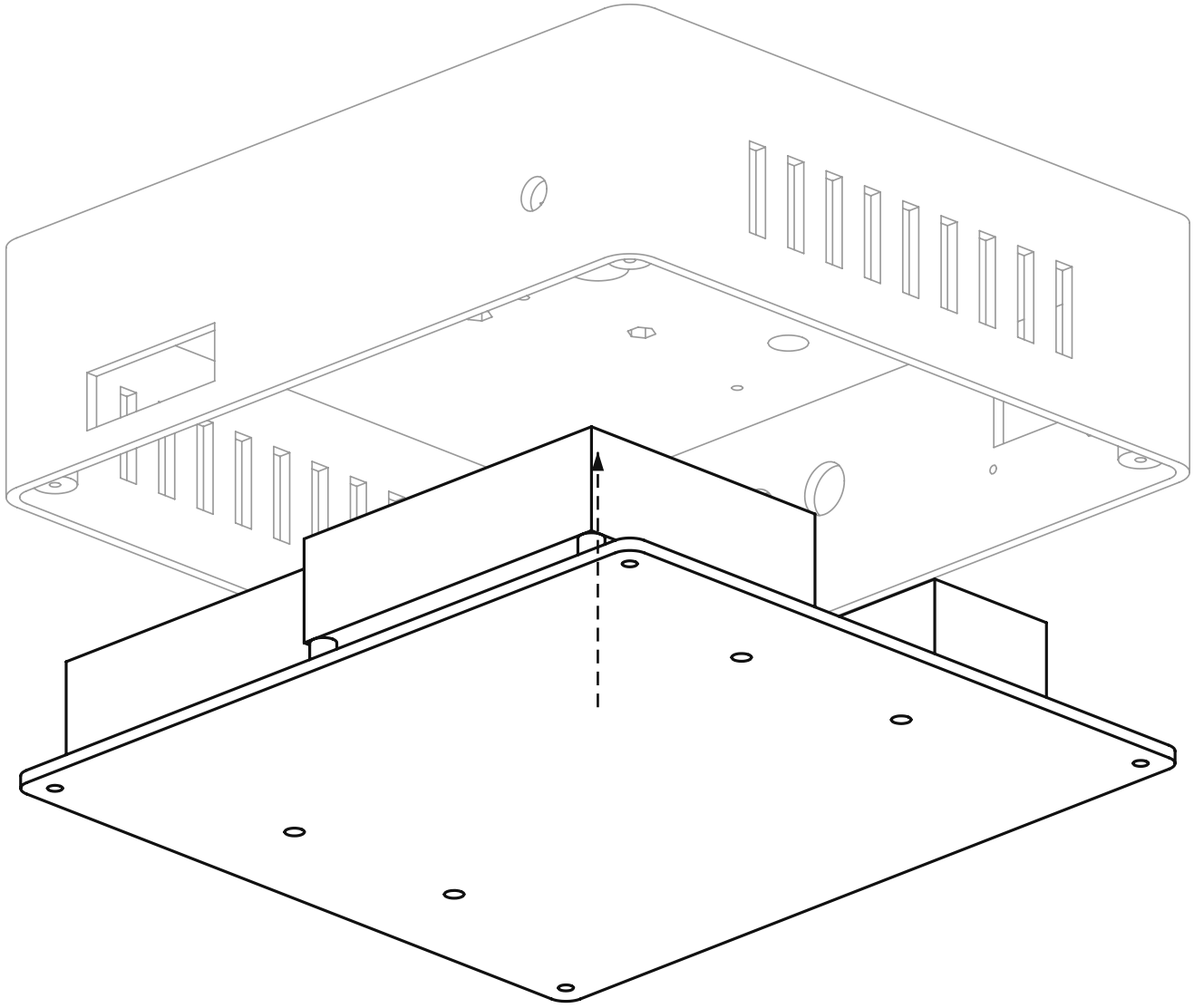




18 Base: close it

OLED, 3 buttons, DC jack and switch into the shell. Wire up (see the electronics sheets), then screw the floor on.

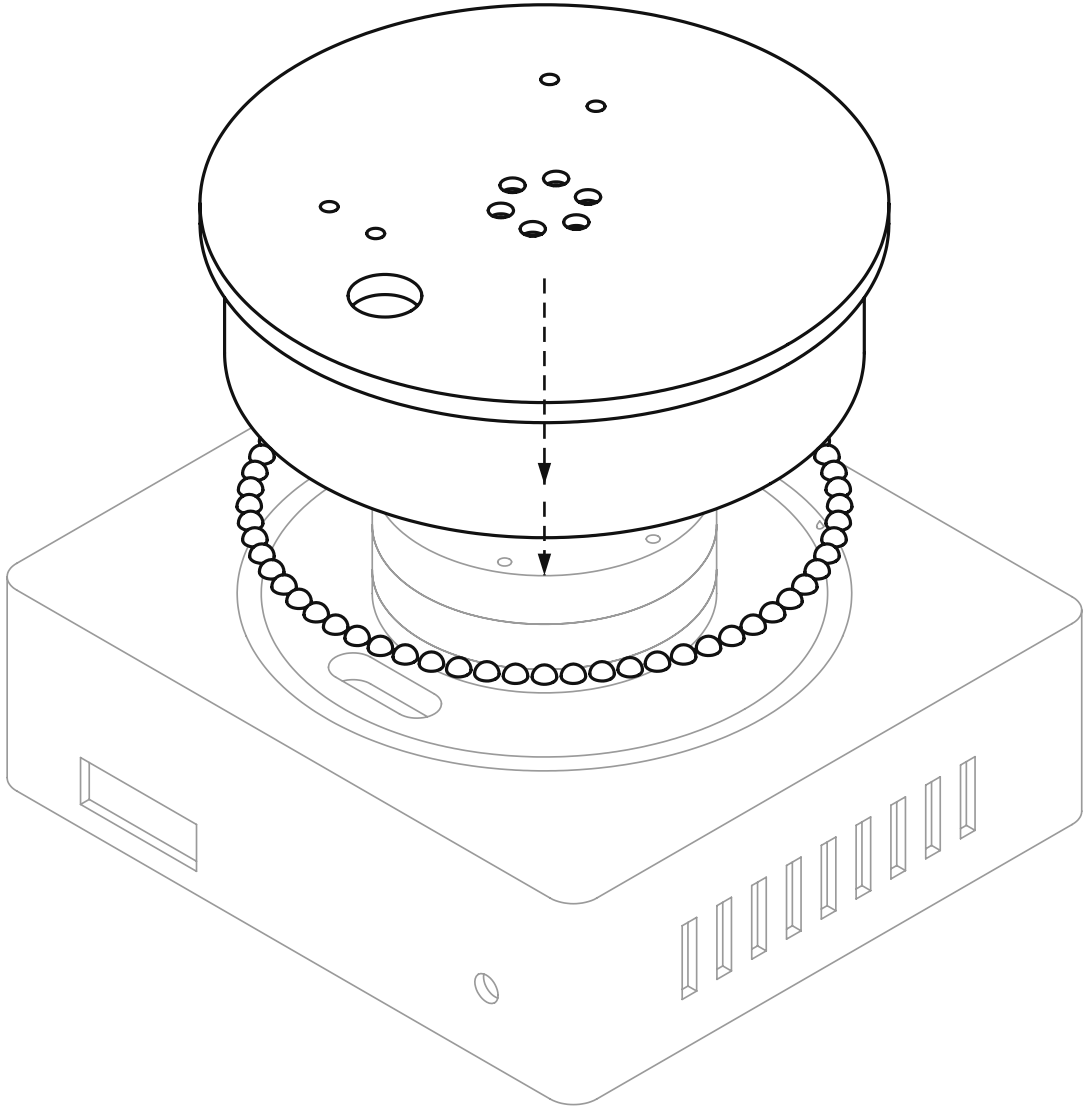




19 Turntable

Grease the race, lay in the 64 balls, set the turntable on the J1 clamp.
Screws go through turntable and clamp into the drive's hub.

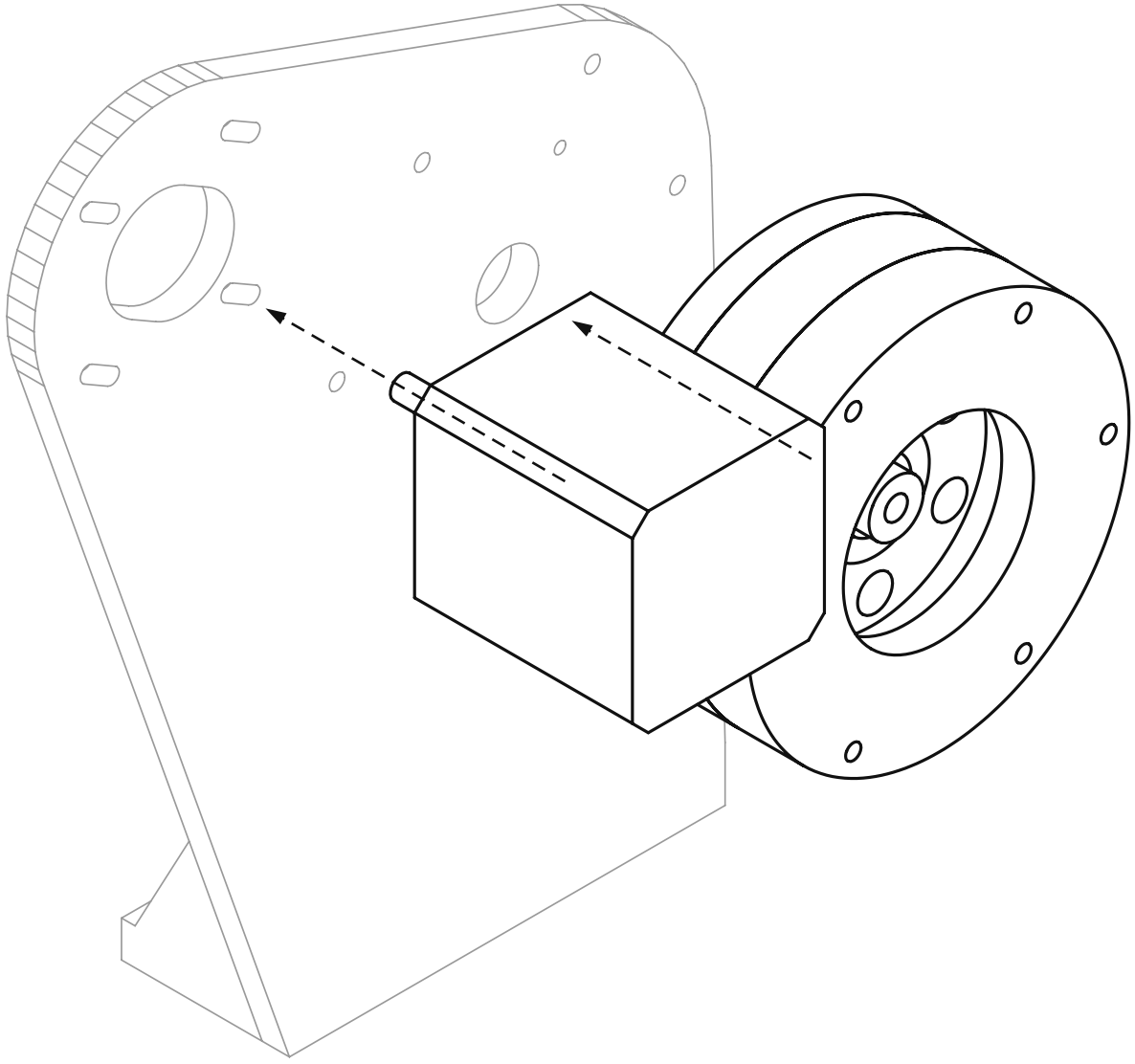




20 Tower: J2 drive and motor

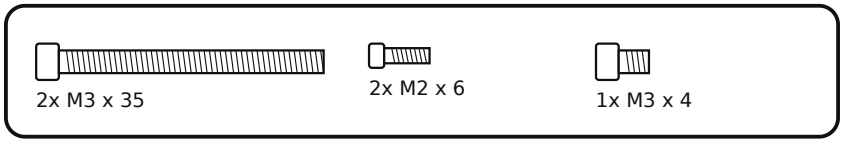
Drive against the tower plate (6 bolts, nuts on the far side).
Motor on the drive side, shaft through the slotted holes.

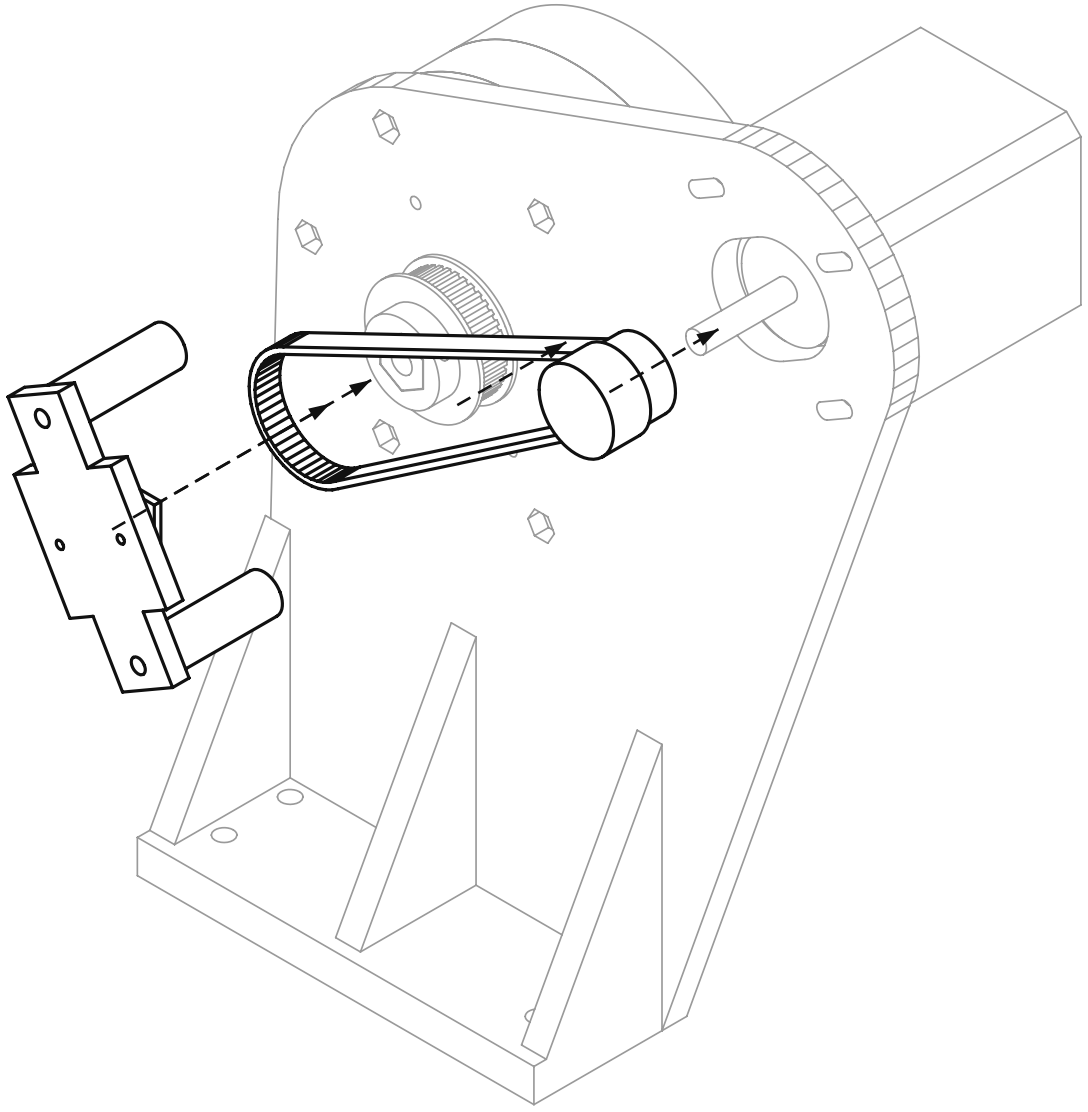




21 Tower: J2 belt and sensor

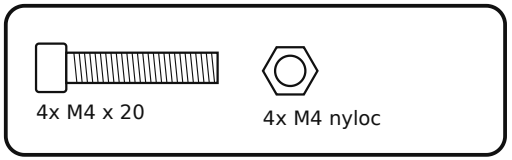
On the back of the tower: pulley, magnet cap, belt, then the sensor bridge. Slide the motor to tension the belt, then tighten.

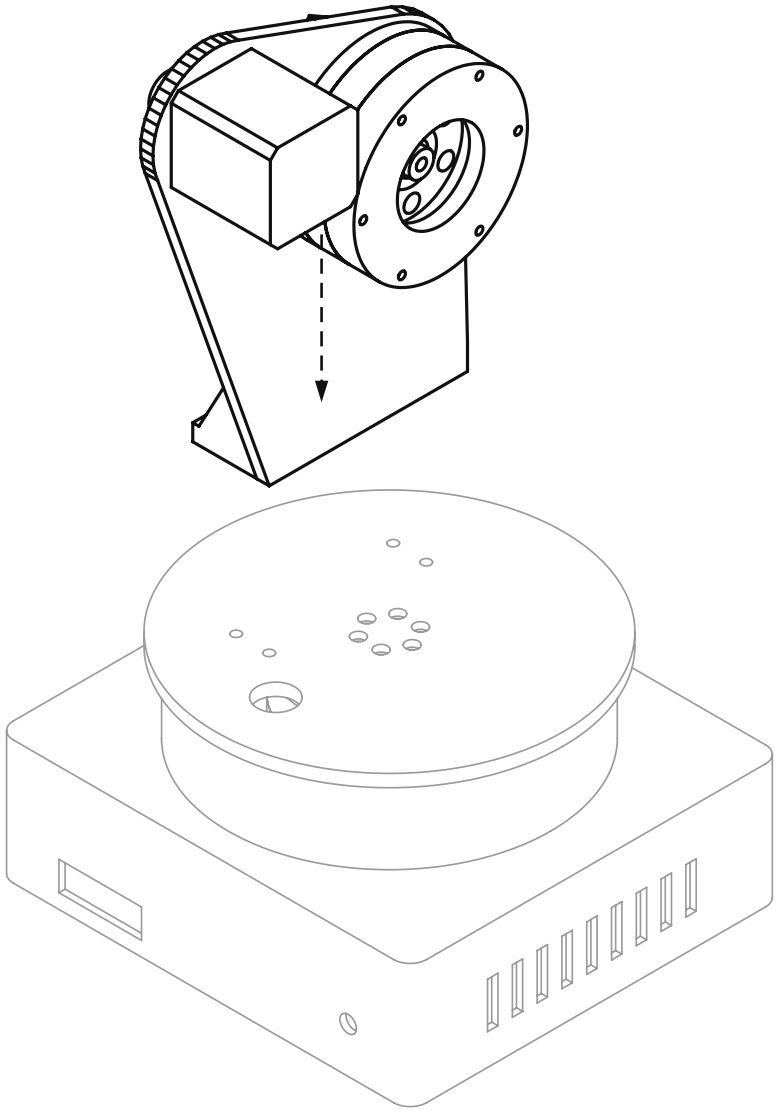




22 Tower onto the turntable

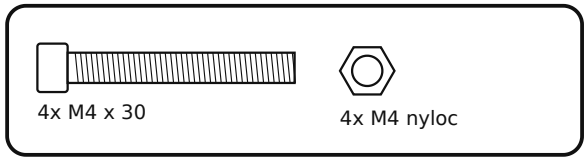
Four M4 bolts through the tower foot and the turntable.

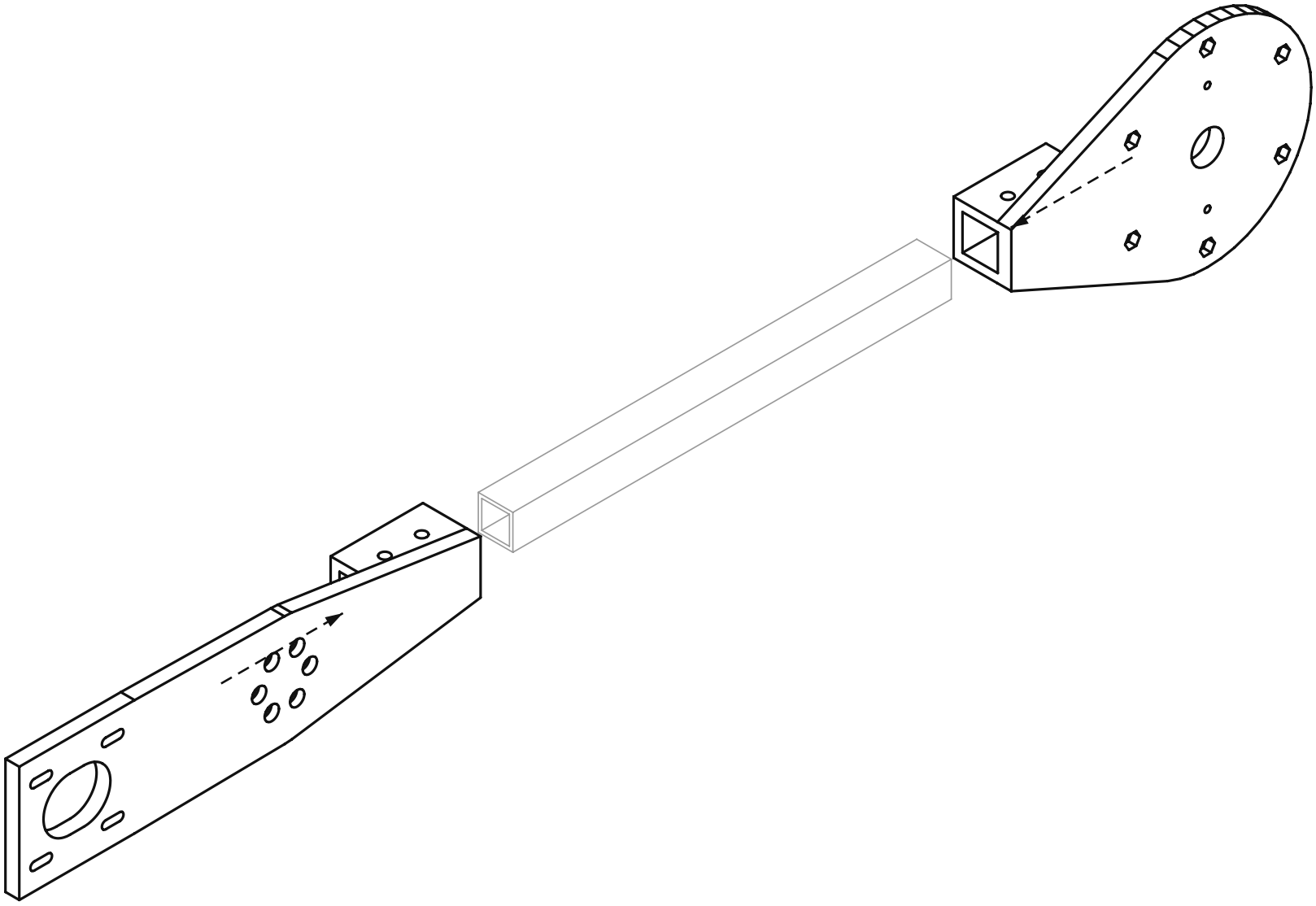




23 Upper arm: tube and fittings

Cut the 15x15x1.5 tube to 190 mm. Push it into both fittings, then bolt across each socket.

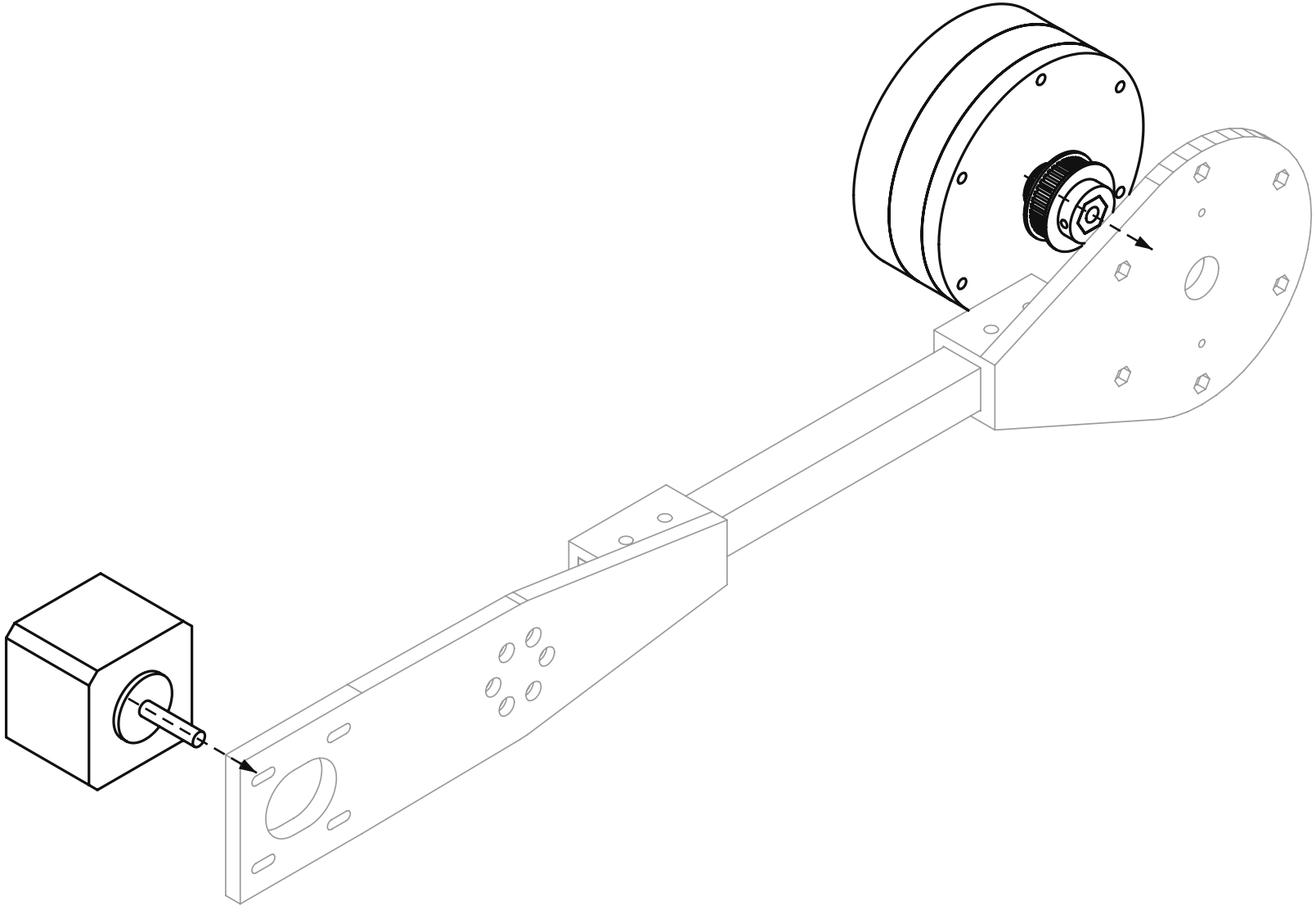




24 Upper arm: J3 drive and motor

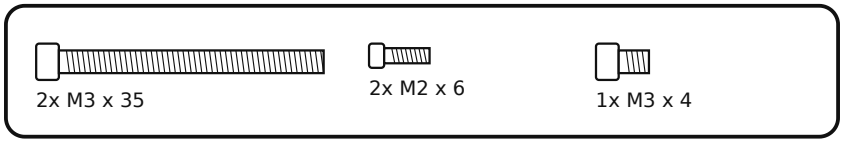
Drive under the elbow plate, motor under the tail plate.
Belt side stays on top.

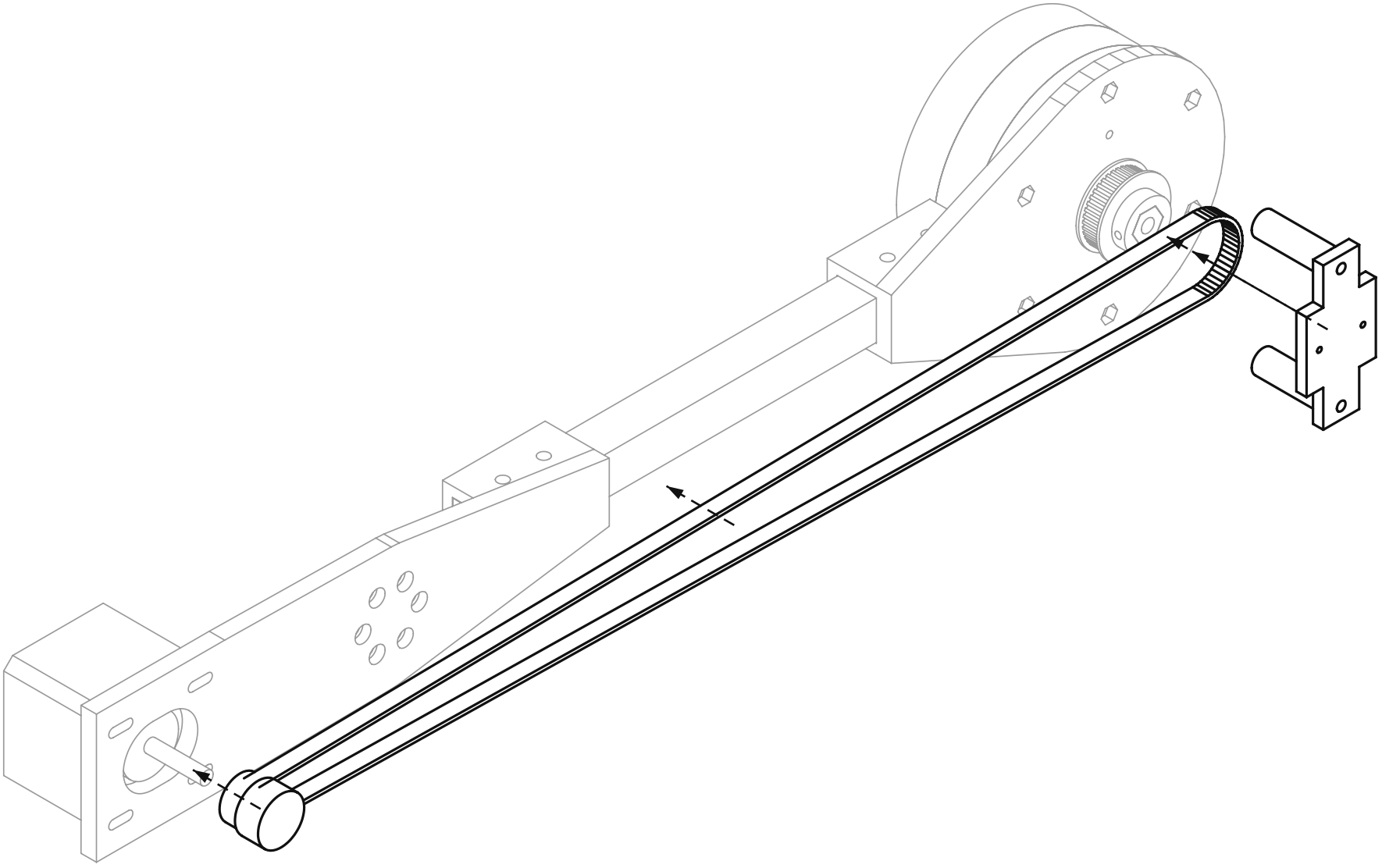




25 Upper arm: J3 belt and sensor

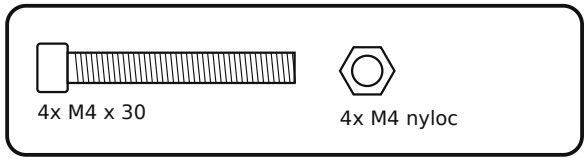
Pulleys, the long belt, magnet cap, sensor bridge.
Slide the motor along its slots to tension the belt.

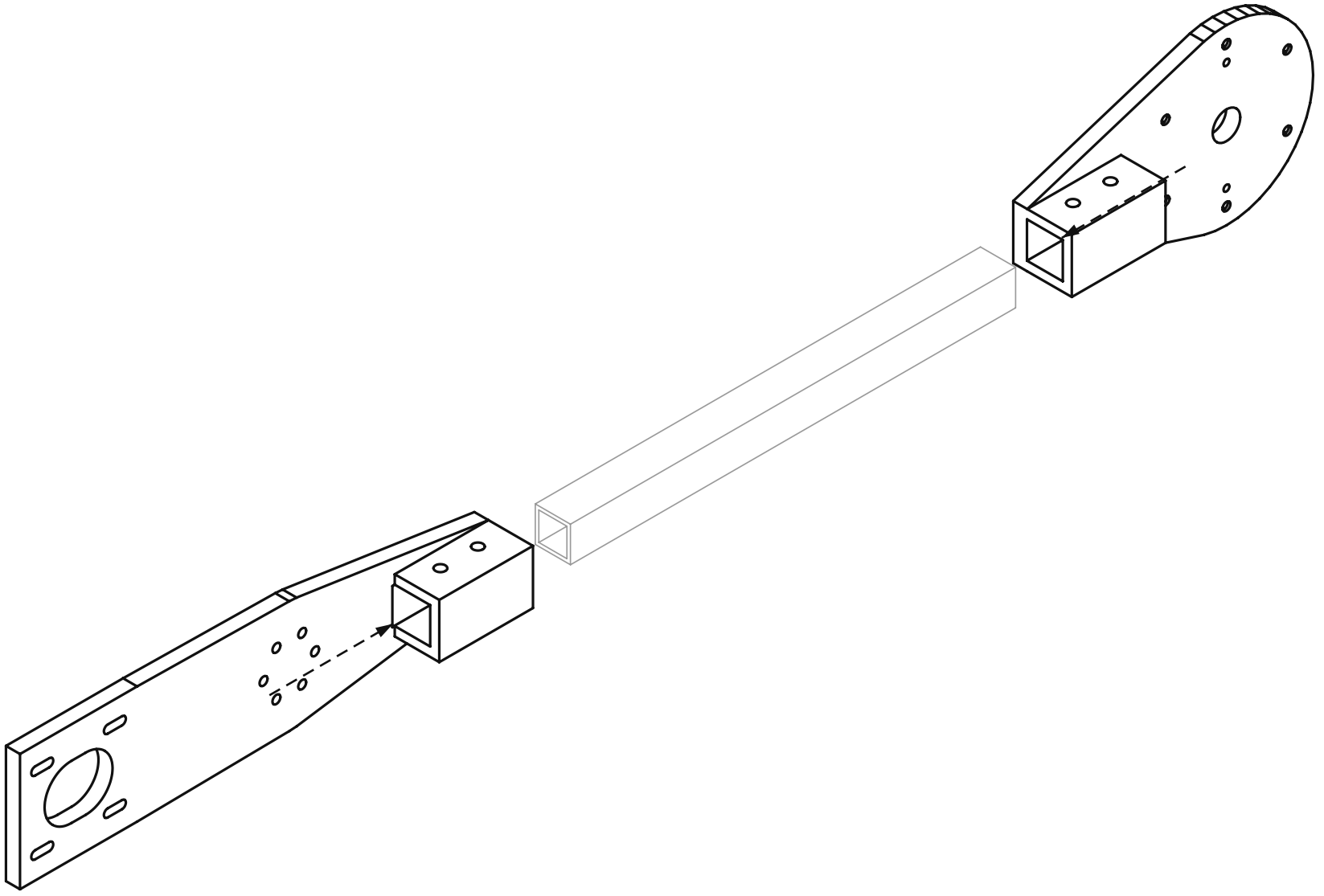




26 Forearm: tube and fittings

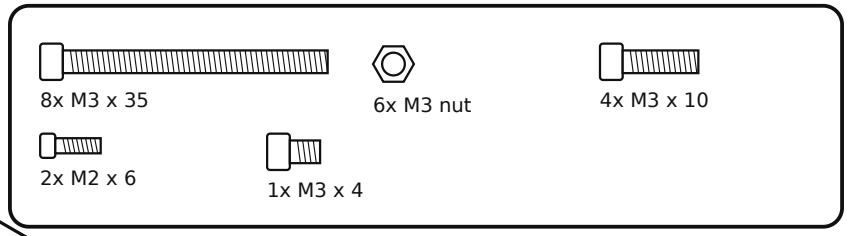
Same as the upper arm: 190 mm tube, 2 bolts per socket.

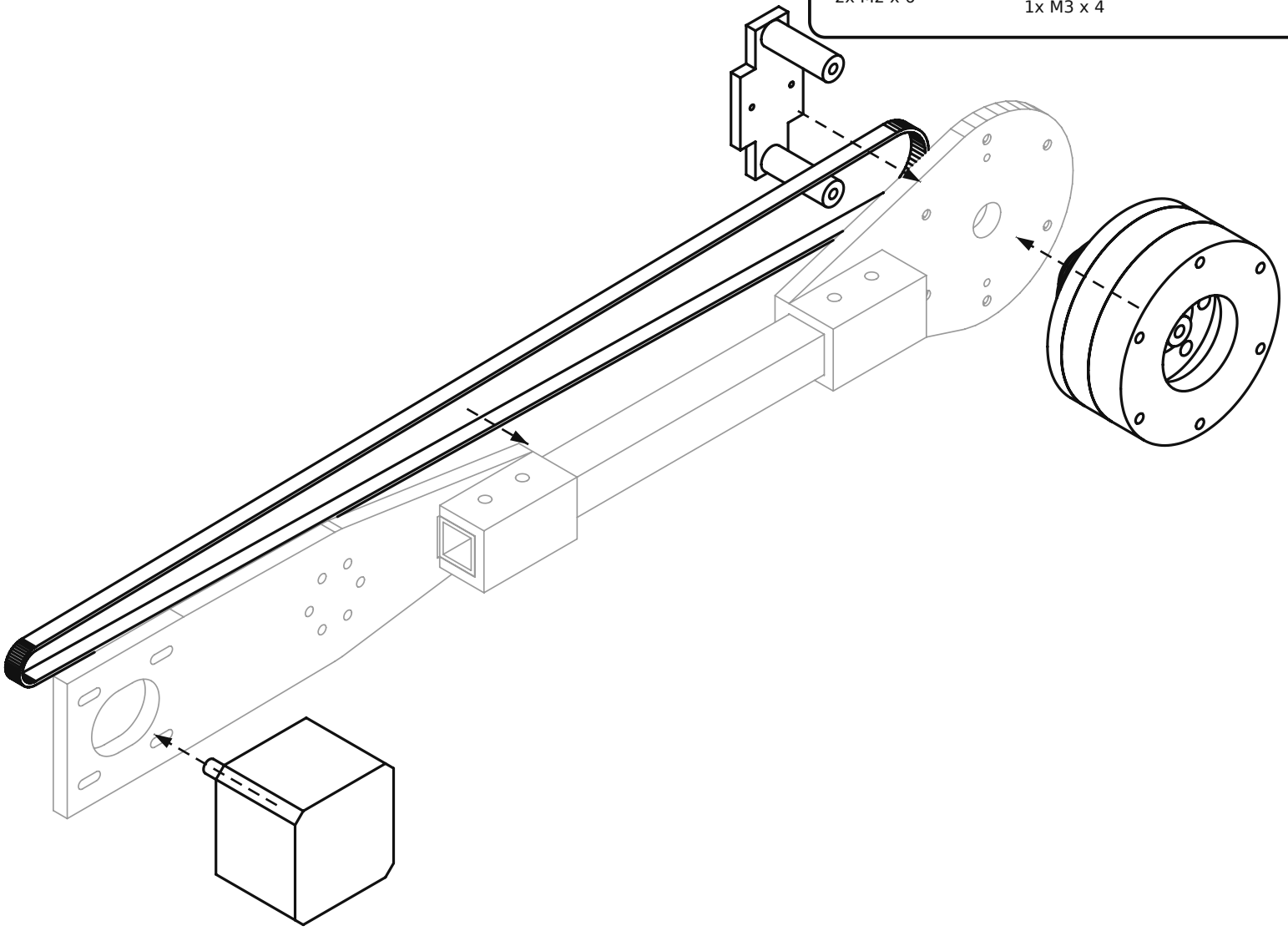




27 Forearm: J4 drive, motor, belt, sensor

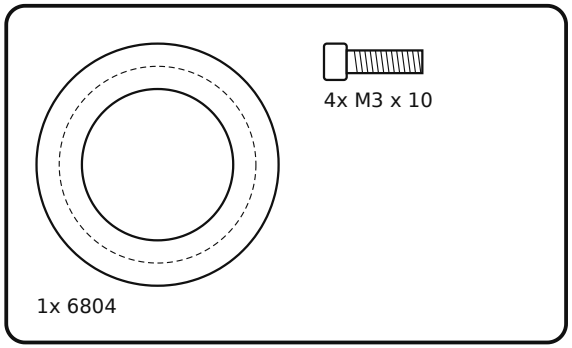
Small drive at the wrist end, motor on the tail.
Belt, pulleys and sensor bridge on the other side of the plate.

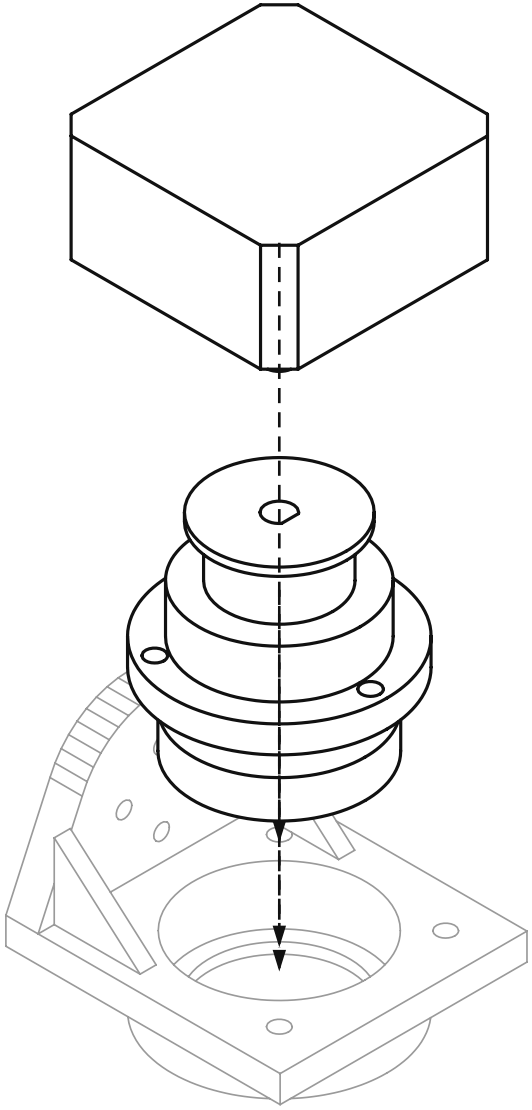




28 Wrist: roll bearing, hub, roll motor

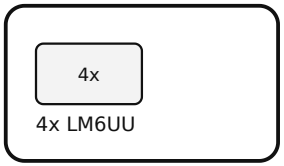
Bearing into the shelf pocket from above, hub through it, then the pancake motor on top: its D-shaft slides into the hub.

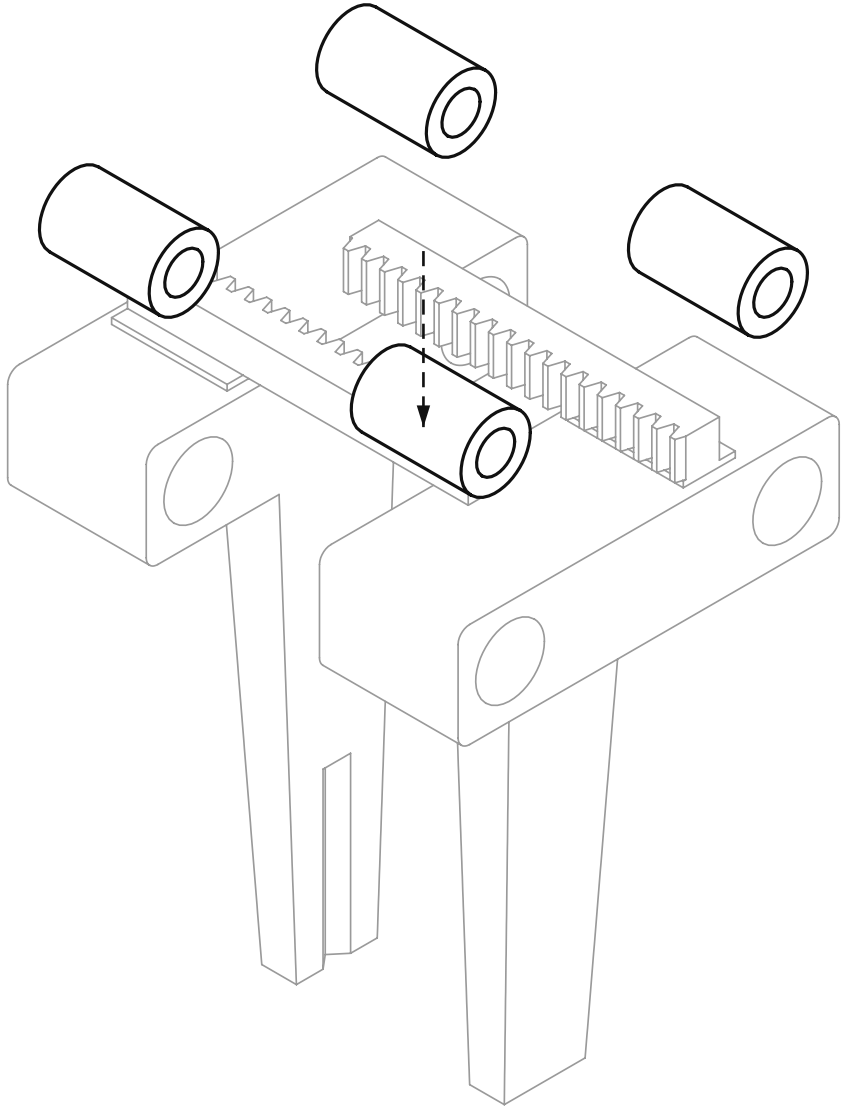




29 Gripper: jaws

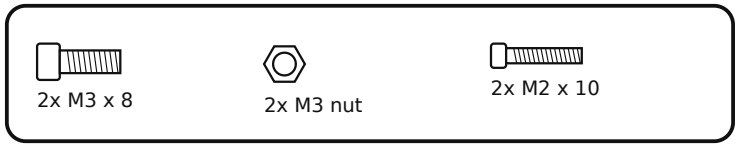
Press two LM6UU bushings into each jaw.

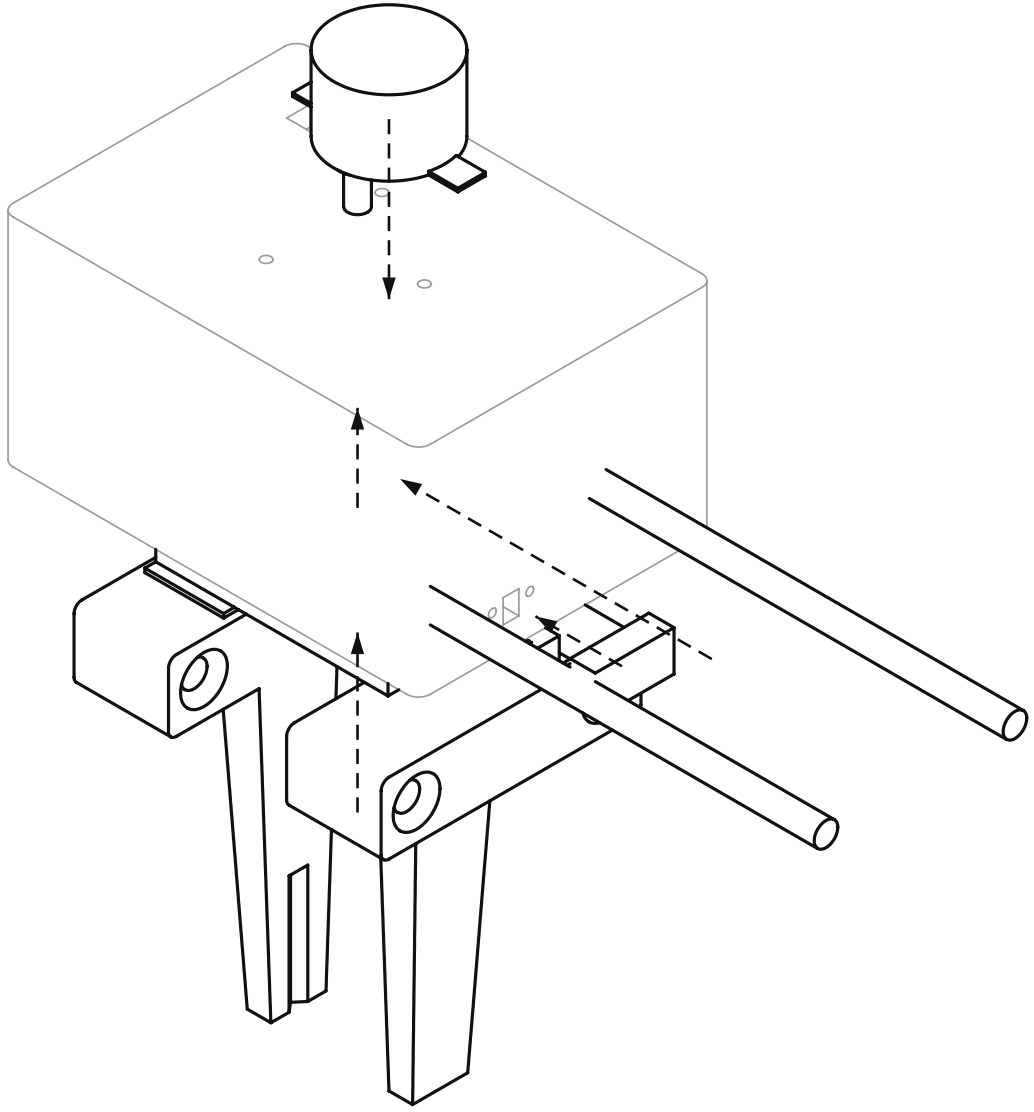




30 Gripper: rods, motor, pinion

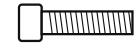
Rods through the end wall and both jaws. Motor on the plate, pinion on its shaft with both jaws at full open. Home switch on the end wall.



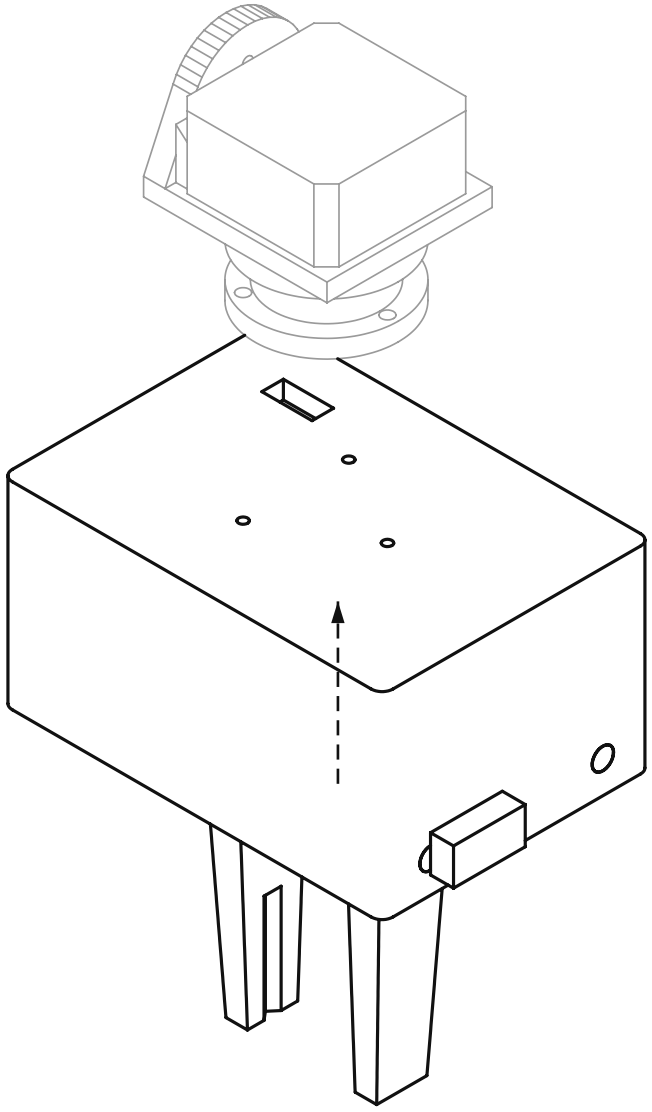


31 Gripper onto the wrist

Three screws through the hub flange into the gripper's top plate.

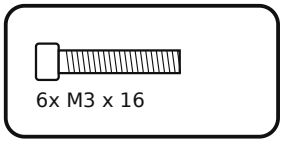


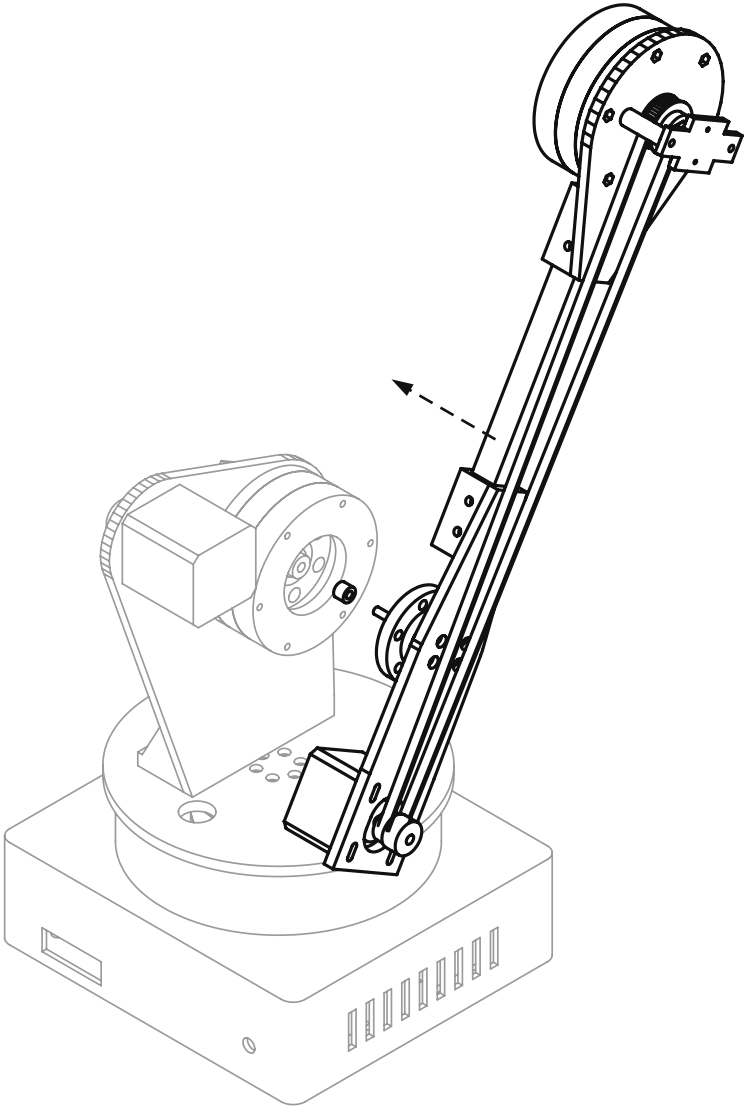
3x M3 x 12



32 Upper arm onto J2

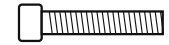
Upper arm's round plate onto the J2 clamp: six screws through plate and clamp into the J2 hub.



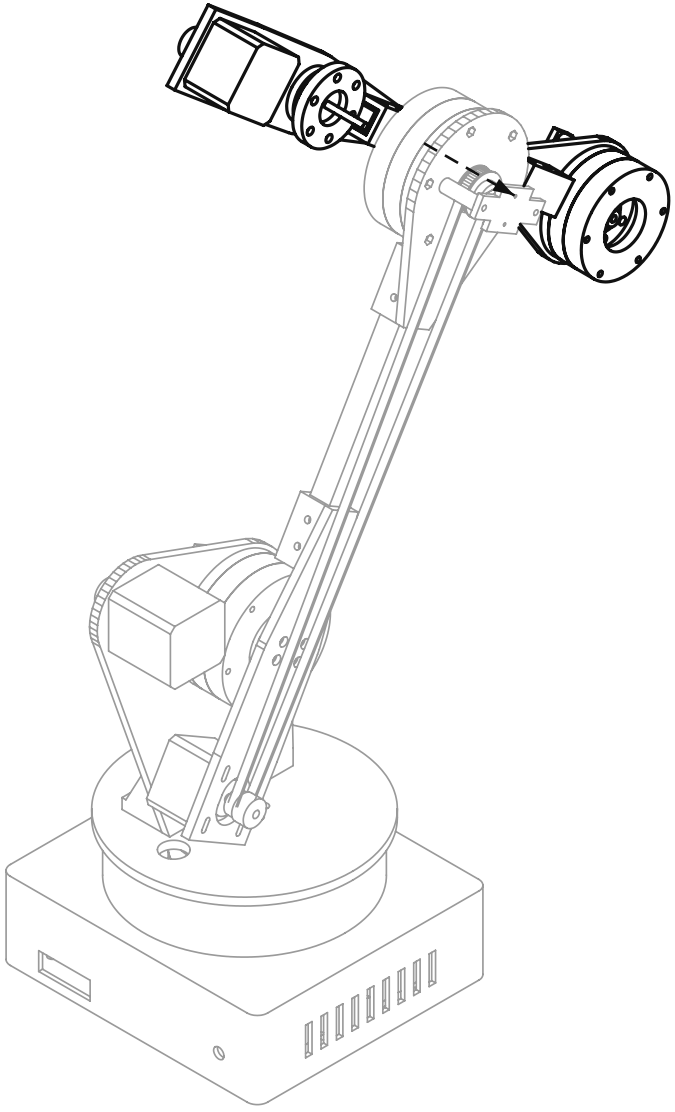


33 Forearm onto J3

Forearm's round plate onto the J3 clamp, six screws.

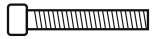


6x M3 x 16

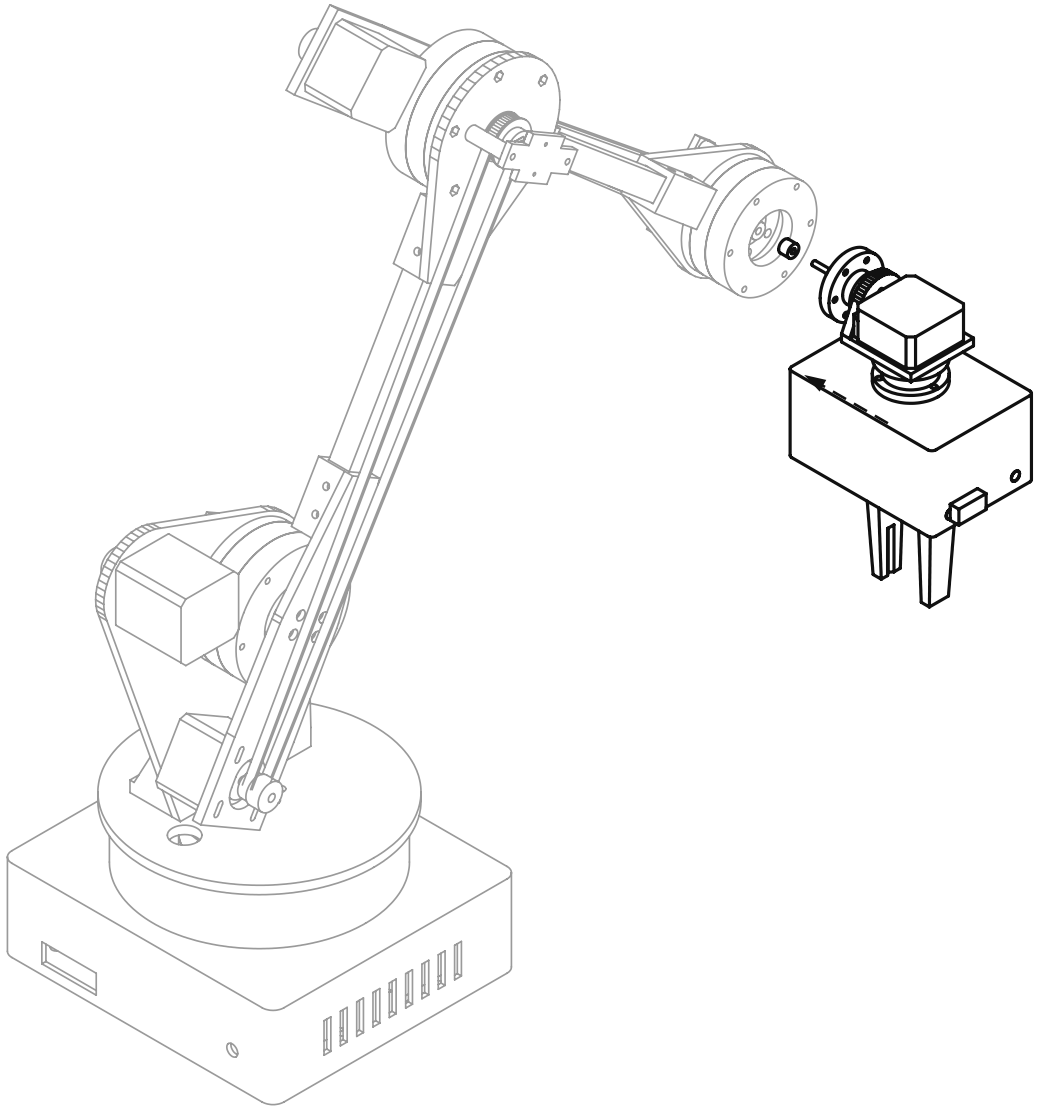


34 Wrist onto J4

Wrist bracket onto the J4 clamp, four M2.5 screws into the J4 hub.

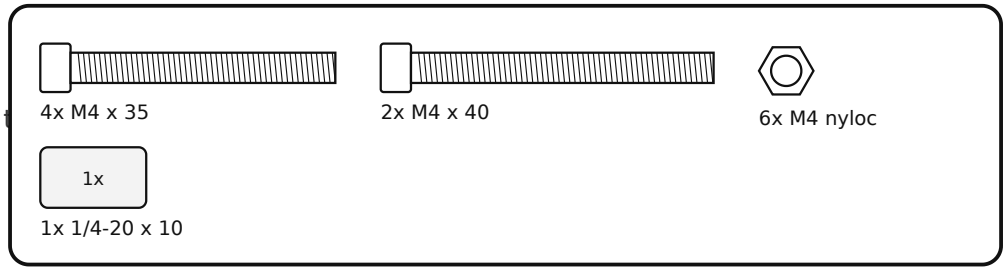


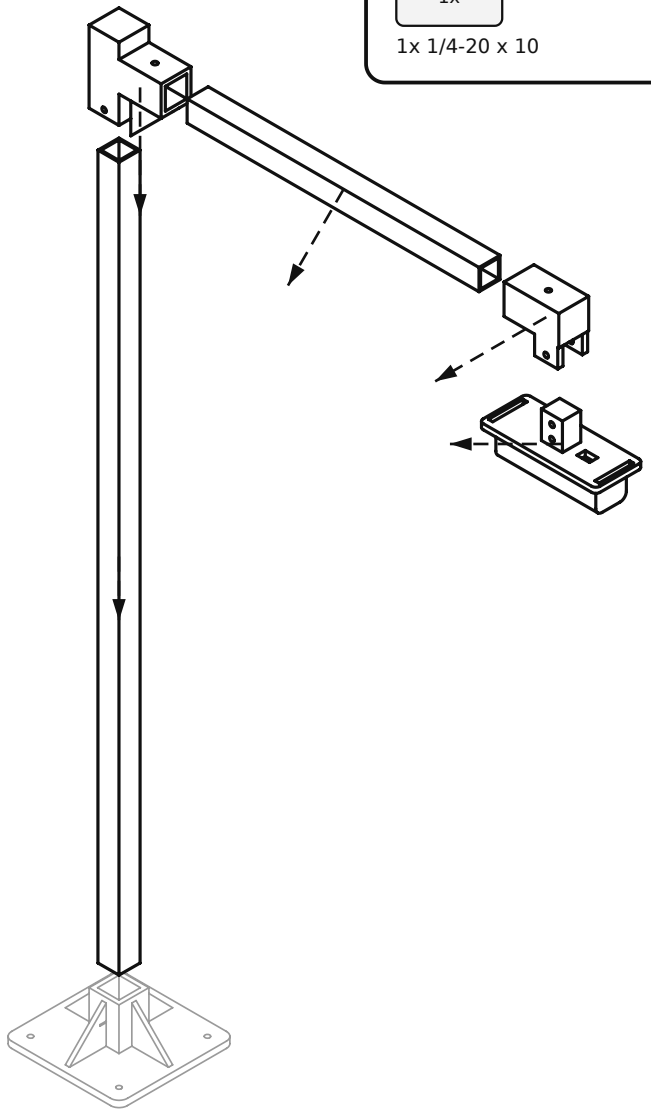
4x M2.5 x 16



35 Camera pole

Cut the 20 mm tube to 669 (pole) and 277 mm (boom).
Foot, pole, tee, boom, mount, camera plate. Level the camera with t





36 Baseplate

Arm base, chessboard and camera pole on one plywood plate. Board edge 100 mm from the base axis, centred; pole at X -300, Y 300. Bolt base and pole foot down.

